CS XI

Computer Science XI — 30-Day Ultimate Cheat-Sheet Booklet

Complete Edition · Days 1–30 · Urdish (Roman Urdu + English) · New Sindh Curriculum 2026

Is booklet mein

1	Computer Systems	Days 1–5 · p. 3
2	Computational Thinking & Algorithm	Days 6–8 · p. 8
3	Programming Fundamentals	Days 9–13 · p. 11
4	Data and Analysis	Days 14–16 · p. 16
5	Application & Impacts of Computing	Days 17–19 · p. 19
6	Digital Literacy	Days 20–23 · p. 22
★	Golden Blitz & Recaps	Days 24–27 · p. 26
M	Mock, Mark & Repair	Days 28–30 · p. 30

30-din ka naqsha (har din = 3 ghante)

Day 01 Teach	Day 02 Teach	Day 03 Teach	Day 04 Teach	Day 05 Checkpoint	Day 06 Teach
Day 07 Teach	Day 08 Checkpoint	Day 09 Teach	Day 10 Teach	Day 11 Teach	Day 12 Teach
Day 13 Checkpoint	Day 14 Teach	Day 15 Teach	Day 16 Checkpoint	Day 17 Teach	Day 18 Teach
Day 19 Checkpoint	Day 20 Teach	Day 21 Teach	Day 22 Teach	Day 23 Checkpoint	Day 24 Golden blitz
Day 25 Golden blitz	Day 26 Golden blitz	Day 27 Recap	Day 28 Mock	Day 29 Mark	Day 30 Repair

Boxes ka matlab

Def (paper mein yehi likho): *Woh definition jo paper mein bilkul aise hi likhni hai.*

🧠 **Yaad rakho:** Mnemonic / trick — jaldi yaad karne ke liye.

📝 **Paper mein:** Board mein kaise poocha jata hai aur poore marks kaise.

⚠ **Dhoka:** Aam ghalti jis se marks katte hain.

★ **Golden** = sab se zyada aane wale 45 topics — pehle yeh. Har din ke aakhir mein **closed-book drill** + jawab.

Naam: _____ Class / Section: _____ Roll no.: _____

Book
XI

Fehrist — Table of Contents

33 cheat-sheet pages · 6 chapters · Golden blitz · Mock paper & key

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30-din progress tracker — har din ke baad bharo

Day	Date	Drill	Done
Day 01		/ 4	☐
Day 02		/ 4	☐
Day 03		/ 4	☐
Day 04		/ 4	☐
Day 05		/ 4	☐
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Day 29		/ 4	☐
Day 30		/ 4	☐

Rozana 3 ghante

0–15′ kal ka ★ recap · 15–105′ aaj ke cards · 105–135′ closed-book drill · 135–180′ ek 5-mark answer.

Print tips

Chrome → Print → A4 · margins **None** · **Background graphics** ON · double-sided theek hai. Mock (Day 28) ke baad wale answer pages alag rakho.

Day 01

Signals, Digital Systems & Boolean Algebra

Discrete & Continuous · Digital Systems · Analog & Digital Signals · Boolean Algebra

★ **Must lock:** Analog and Digital Signals

1.1.1 Discrete vs Continuous Quantities

Jo cheez **gin** sakte ho ya **naap** sakte ho — woh *quantity* hai. Do qisam:

Discrete (gina jata hai)	Continuous (naapa jata hai)
Class mein students ki tadaad	Pani ka temperature
Parking mein cars	Chalti car ki speed
Fixed, alag alag values (1,2,3 — beech mein kuch nahi)	Range ke andar **koi bhi** value



🧠 **Yaad rakho:** Seerhi (stairs) pe sirf step 1, 2, 3 pe khare ho sakte ho = discrete. Ramp pe kisi bhi height pe = continuous.

1.1.2 Digital Systems

Def (paper mein yehi likho): *A digital system is an electronic system that uses only two values: 0 and 1.*

0 = OFF / LOW / FALSE · **1** = ON / HIGH / TRUE. Ek 0 ya 1 ko **bit** kehte hain — jaise light switch.

Faida (Advantage)	Matlab
Reliable	Noise / interference ka asar kam
Accurate	Copy karne pe quality loss nahi
Easy to design	Sirf do states (0, 1) handle karni hain
Programmable	Software se aasani se control

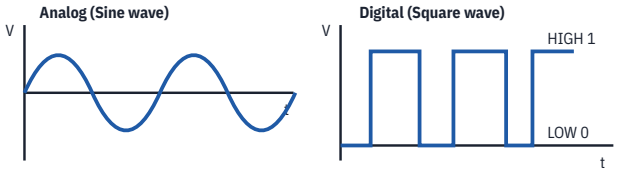
5-mark frame: Analog & Digital Signals

Define	A signal is a changing voltage that carries information; analog is continuous, digital has only two levels.
Explain	Analog smoothly badalta hai (infinite values); digital sirf HIGH(1)/LOW(0) pe jump karta hai.
Example	Analog: human voice, mercury thermometer. Digital: computer data.
Diagram	Sine wave (V–t) + square wave (HIGH/LOW) — dono axes label.
Working	Comparison table: nature, shape, reliability (noise), examples.

1.1.3 Analog and Digital Signals ★ Golden

Def (paper mein yehi likho): *A signal is a changing voltage that carries information.*

Analog = continuous, waqt ke saath *smoothly* badalta hai (sine wave) — insani awaaz, mercury thermometer.
Digital = sirf do levels: HIGH (1) aur LOW (0) — square wave, computer data.



Analog ⟶ smooth curve | Digital ⟶ achanak jumps (sirf 2 levels)

Feature	Analog	Digital
Nature	Continuous flow	Discrete values (bits)
Wave shape	Sine wave (smooth)	Square wave (jumps)
Reliability	Low — noise ka asar	High — noise-resistant
Examples	Human voice, thermometer	Computer data

🧠 **Yaad rakho:** Analog = samundar ki lehrein 🌊. Digital = qila ki deewar ke daant (square teeth) 🏰.

📝 **Paper mein:** "Differentiate analog & digital signals" (3–4 marks) + "Draw both". Table mein kam az kam 4 points + dono graph label ke saath (V, t, HIGH/LOW).

1.1.4 Boolean Algebra

George Boole (English mathematician) ne banaya. Sirf do values: TRUE (1) aur FALSE (0). Har program ki `if` condition chhota sa Boolean algebra hai.

Concept	Matlab	Example
Variables	Letters jin mein 0 ya 1 ho	A, B, C, X, Y
Operations	Variables pe actions	AND, OR, NOT
Expressions	Variables + operations	$Y = A + B$

⚠ **Dhoka:** Boolean mein "+" ka matlab **OR** hai, addition nahi!

$$1 + 1 = 1$$

(OR), 2 nahi.

Closed-book drill: (1) Kitaab ke pages discrete ya continuous? (2) Digital mein 1 kya represent karta hai? (3) Digital signal ki wave shape? (4) Boolean algebra kisne banaya?

Jawab: 1. Discrete (gin sakte ho) · 2. ON/HIGH/TRUE · 3. Square wave · 4. George Boole | **Aaj raat ka 5-mark:** Analog & Digital Signals — Define · Explain · Example · Diagram · Working.

Boolean Operations, Truth Tables & Logic Gates

Boolean Operations · Truth Tables · Basic Gates · Universal Gates (NAND, NOR)

★ **Must lock:** Logic Gates (AND, OR, NOT)

1.1.5 Boolean Operations

Teen basic operations — baqi sab inhi se bante hain.

Operation	Rule	Real-life misaal
AND (·)	Output 1 sirf jab **SAB** inputs 1	Darwaze pe 2 taale — dono khulein tab darwaza khule
OR (+)	Output 1 jab **koi ek** bhi input 1	Kamre ke 2 darwaze — koi bhi khula ho, andar ja sakte ho
NOT (')	Ulta kar deta hai	Light switch — ON ho to roshni, OFF ho to andhera

🧠 **Yaad rakho:** AND = "strict" (sab ko 1 hona hai). OR = "easy-going" (ek 1 kaafi). NOT = "baaghi" (hamesha ulta).

1.1.6 Truth Tables

Def (paper mein yehi likho): *A truth table shows all possible input combinations and the output for each one.*

1. Inputs gino → n
2. Total rows $= 2^n$ ($n = 2 \Rightarrow 4$, $n = 3 \Rightarrow 8$, $n = 4 \Rightarrow 16$)
3. Combinations binary counting order mein likho (0,1,2,3...)
4. Har row ka output nikalo

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

🧠 **Yaad rakho:** 3-input fill trick: **C** har row badlo (0,1,0,1...), **B** har 2 rows (0,0,1,1...), **A** har 4 rows (0,0,0,0,1,1,1,1).

Bonus: Basic Boolean Laws

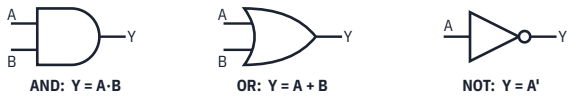
Law	AND form	OR form
Identity	$A \cdot 1 = A$	$A + 0 = A$
Null	$A \cdot 0 = 0$	$A + 1 = 1$
Idempotent	$A \cdot A = A$	$A + A = A$
Complement	$A \cdot \bar{A} = 0$	$A + \bar{A} = 1$
Double NOT	$\overline{\bar{A}} = A$	
De Morgan	$\overline{A \cdot B} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A} \cdot \bar{B}$

🧠 **Yaad rakho:** De Morgan: "bar todo, sign badlo" — NAND = OR of inverted inputs; NOR = AND of inverted inputs.

1.1.7 Logic Gates (AND, OR, NOT) ★ Golden

Def (paper mein yehi likho): *Logic gates are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit.*

Har digital system ki **building blocks** — ek processor mein arbon (billions) gates. NOT gate ko **inverter** bhi kehte hain.



AND = "D" shape (flat back) · OR = curved back + nokeela front · NOT = triangle + bubble

A	B	AND	OR	NOT A
0	0	0	**0**	1
0	1	0	1	1
1	0	0	1	0
1	1	**1**	1	0

🧠 **Yaad rakho:** Sirf "special row" yaad karo: AND = 1 sirf **11** pe. OR = 0 sirf **00** pe.

📝 **Paper mein:** "Draw symbol, write expression & truth table" (2–3 marks har gate). Inputs A, B aur output Y zaroor label karo.

1.1.7 Universal Gates (NAND, NOR)

NAND = NOT AND: $Y = \overline{A \cdot B}$

| **NOR** = NOT OR: $Y = \overline{A + B}$

. Output pe chhota circle (**bubble**) = NOT.



A	B	NAND	NOR
0	0	1	**1**
0	1	1	0
1	0	1	0
1	1	**0**	0

Universal kyun? Duniya ka koi bhi logic circuit sirf NAND (ya sirf NOR) gates se ban sakta hai.

🧠 **Yaad rakho:** NAND = AND ka bilkul ulta (0 sirf 11 pe). NOR = OR ka ulta (1 sirf 00 pe). Bas column flip karo.

Drill: (1) OR ka symbol? (2) 4 inputs → kitni rows? (3) OR gate, A=0, B=0 → output? (4) NAND/NOR universal kyun?

Jawab: 1. "+" · 2. $2^4 = 16$ · 3. 0 · 4. Koi bhi circuit sirf NAND ya sirf NOR se ban sakta hai | **5-mark:** Logic Gates (AND, OR, NOT).

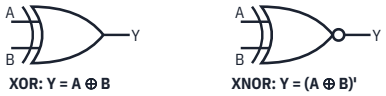
1.1.7 XOR & XNOR

XOR (Exclusive OR): output 1 jab inputs **mukhtalif**.

$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

XNOR = XOR + NOT: output 1 jab inputs **aik jaise** ("equality gate").

$$Y = \overline{A \oplus B}$$



A	B	XOR	XNOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

XOR = OR symbol + peeche extra curve · XNOR = XOR + bubble

1.1.8–1.1.10 Expressions, Minterms, Maxterms

Order of precedence (BODMAS ki tarah):

1. ( ) Brackets → 2. NOT ' → 3. AND · → 4. OR +

Example

$$Y = A + \bar{B}C$$

: pehle

$$\bar{B}$$

, phir

$$\bar{B} \cdot C$$

, aakhir mein

$$A + (\dots)$$

.

	Minterm (SOP)	Maxterm (POS)
Kahan se?	Rows jahan output **1**	Rows jahan output **0**
Rule	0 → bar, 1 → plain (AND)	1 → bar, 0 → plain (OR)
A=0,B=1,C=1	$\bar{A}\,B\,C$	$A + \bar{B} + \bar{C}$
Form	$F = \sum m(\dots)$	$F = \prod M(\dots)$

Worked:

$$Y = A + \bar{B}C$$

#	A	B	C	$\bar{B}$	$\bar{B}C$	Y
0	0	0	0	1	0	0
1	0	0	1	1	1	**1**
2	0	1	0	0	0	0
3	0	1	1	0	0	0
4–7	1	x	x	–	–	**1**

$$\Rightarrow Y = \sum m(1,4,5,6,7) = \prod M(0,2,3)$$

🧠 **Yaad rakho:** SOP = 1s, POS = 0s. "**Min** ko 1 pasand, **Max** ko 0 pasand."

K-Map: 6-step method

1. Variables gino → cells

$$= 2^n$$

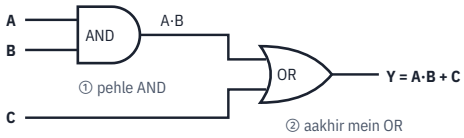
(2-var 4, 3-var 8, 4-var 16)

2. Gray-code headings likho (00, 01, 11, 10)
3. Har minterm ki cell mein 1, baqi 0
4. Sab se bare groups (8 → 4 → 2 → 1) banao, overlap allowed
5. Har group: jo variable **same** rahe woh likho, jo badle woh gira do
6. Groups ko OR (+) se jodo

1.1.11 Logic Diagrams ★ Golden

Def (paper mein yehi likho): *A logic diagram shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression.*

Precedence follow karo: sab se pehle hone wali operation **inputs ke qareeb**, aakhri operation **Y ke paas** (right side).



🧠 **Yaad rakho:** Poochho "sab se aakhir mein kya hota hai?" — woh gate bilkul right pe Y ke saath. Phir peeche inputs ki taraf chalo.

$$Y = A \cdot B + C$$

= 2 gates.

📝 **Paper mein:** "Draw logic diagram of Y = ..." + kabhi "truth table bhi banao" (4–5 marks). Pehle precedence steps ek line mein, phir saaf symbols.

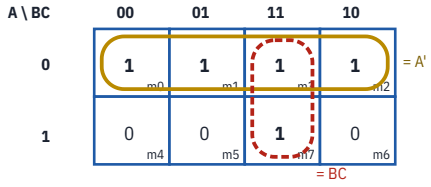
1.1.13 Karnaugh Maps (K-Maps) ★ Golden

Def (paper mein yehi likho): *A K-map is a graphical tool used to simplify Boolean expressions. Adjacent cells differ by only ONE bit (Gray code).*

Gray code order: **00, 01, 11, 10** (har step pe sirf 1 bit badle).

Example:

$$F(A,B,C) = \sum m(0,1,2,3,7)$$



$$\Rightarrow F = \bar{A} + BC$$

(4 ka group →

$$\bar{A}$$

; 2 ka group →

$$BC$$

)

- Groups sirf **1, 2, 4, 8** (power of 2), rectangle shape, jitne bare utna acha.
- Left-right edges jurre hue hain (map ek **tube** pe hai) — wrap-around allowed.
- Har group mein jo variable badalta hai, woh **gir** jata hai.

📝 **Paper mein:** Favourite long question (5–6 marks): "Simplify F(A,B,C) = Σm(...) using K-map." Gray-code headings, har cell bharo (0s bhi), groups circle karo, final expression.

Layouts yaad rakho

Type	Rows	Columns
2-var (4 cells)	A = 0, 1	B = 0, 1
3-var (8 cells)	A = 0, 1	BC = 00, 01, 11, 10
4-var (16 cells)	AB = 00, 01, 11, 10	CD = 00, 01, 11, 10

⚠ **Dhoka:** Columns 00, 01, **10**, 11 likhna ghalat hai — Gray code mein 11 pehle, 10 baad mein!

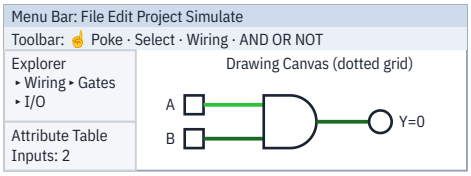
LogiSim, SDLC & Waterfall Model

LogiSim Evolution · Building a Circuit · Software Development Life Cycle · Waterfall

★ **Must lock:** SDLC, Waterfall Model

LogiSim Evolution — Simulation Tool

Free graphical tool jis mein logic circuits **design** aur **simulate** karte hain. "Simulate" = computer asli circuit ban jata hai: inputs on/off karo, output badalta dekho.



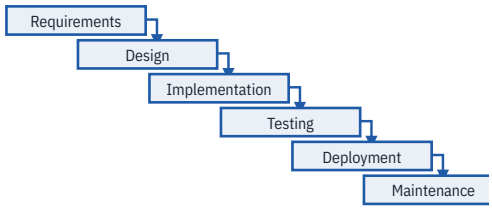
Area	Kaam
Menu Bar	File, Edit, Project, Simulate
Toolbar	Wiring, Poke, AND/OR/NOT
Explorer (left)	Libraries: Gates, I/O, Wiring
Attribute Table	Gate properties: **No. of Inputs**, Facing
Drawing Canvas	Dotted grid — gates rakho, wire karo

🧠 **Yaad rakho:** **Bright green** = 1, **dark green** = 0, **blue** = wire connect nahi, **red** = error. Input pin = **square**, output pin = **circle**. Poke = ungli 👆.

Y = A·B banao: AND gate rakho → left 2 input pins, right 1 output pin → Wiring tool se jodo → Poke se A, B badlo. Sirf A=1, B=1 pe output wire **bright green** (Y=1).

1.2.4 The Waterfall Model ★ Golden

Def (paper mein yehi likho): *The Waterfall model is a linear, sequential approach to the SDLC; each phase must be completed before the next begins.*



Pros ✔	Cons ✘
Samajhna, manage karna aasaan	Changes ke liye flexible nahi
Har step pe documentation	Testing bohat der se

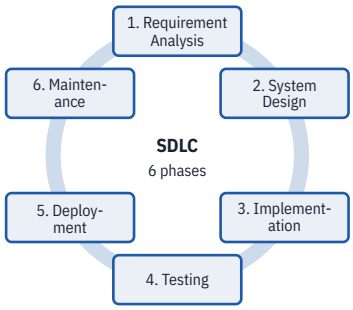
Kab use karein? Requirements clear & fixed, project chhota/stable, technology jaani-pehchani.

🧠 **Yaad rakho:** Pani sirf neeche girta hai — wapas upar nahi chadhta.

1.2.1–1.2.3 SDLC ★ Golden

Def (paper mein yehi likho): *The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.*

Faide: clear goals, behtar planning, risk kam, high quality. Iske baghair: features bhool jaate, budget over, buggy software.



Phase	Kya hota hai	Deliverable
1. Requirement Analysis	Client se milo, zaroorat samjho, features list. **Sab se ahem** — yahan ki ghalti har phase mein phailti hai	**SRS** document
2. System Design	High-Level Design (architecture), Low-Level Design (DB, UI), flowcharts	Design Document
3. Implementation	Asal code (Python, Java...)	Working source code
4. Testing	Bugs dhoondo — **black box** (andar nahi dekhte) / **white box** (code dekh ke)	Test report
5. Deployment	User ko software dena / install	Live system
6. Maintenance	Bug fix, updates, naye features	Updated versions

🧠 **Yaad rakho:** "**R**eal **D**evelopers **I**mplement, **T**est, **D**eploy, **M**aintain". Black box = band dabba; white box = sheeshe ka dabba.

📝 **Paper mein:** "Explain phases of SDLC" (5–8 marks): 6 phases tarteeb se, har ek ki 1–2 line + deliverable.

5-mark frame: SDLC

Define	SDLC is a structured process that guides software development from beginning to end.
Explain	6 phases ek tarteeb mein; har phase ka apna deliverable; ghalti jaldi pakri jati hai.
Example	School result app: requirements (teachers se), design (screens), code, test, install, updates.
Diagram	6 boxes ka cycle (Requirement → ... → Maintenance → wapas).
Working	Har phase ki activity + deliverable (SRS, design doc, code, test report).

Drill: (1) Gate ke inputs kahan badalte ho? (2) Input pin square ya circle? (3) SRS kis phase se? (4) Waterfall "sequential" kyun?

Jawab: 1. Attribute Table · 2. Square · 3. Requirement Analysis · 4. Har phase poora ho tab agla, fixed order | **5-mark:** SDLC.

Agile, Waterfall vs Agile & OSI / TCP-IP

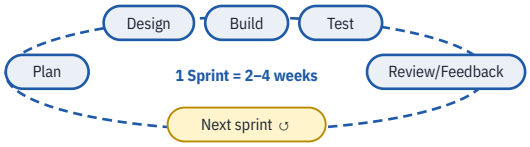
Agile Model · Comparison · Communication Models · OSI 7 Layers · TCP/IP

★ **Must lock:** Agile Model, OSI 7-Layer Model

1.2.4 The Agile Model ★ Golden

Def (paper mein yehi likho): *The Agile model is an iterative approach: software is built in small, repeated cycles called sprints (usually 2–4 weeks).*

Har sprint mein chhota hissa plan → build → test → customer ko dikhao. Customer **feedback continuously** aata hai aur agle sprint mein use hota hai.



Pros ✔	Cons ✘
Changes ke liye highly flexible	Documentation kam
Working software jaldi milta hai	End-time aur cost unpredictable

🧠 **Yaad rakho:** Agile = phurteela athlete 🏃 — jaldi direction badle. Pizza misaal: customer har slice chakhta hai, agla slice uske hisaab se.

📝 **Paper mein:** Ye 3 alfaaz zaroor likho: "**iterative**", "**sprints (2–4 weeks)**", "**continuous customer feedback**".

1.2.5–1.2.7 Waterfall vs Agile

Dono SDLC ke same phases follow karte hain — farq **tareeqe** mein: Waterfall ek dafa seedhi line; Agile chhote loops mein bar bar.

Dimension	Waterfall	Agile
Flexibility	Rigid — baad mein change mushkil	Bohat flexible
Customer	Sirf shuru aur aakhir	Har sprint ke baad (continuous)
Delivery	Ek dafa, aakhir mein	Bar bar, partial deliveries
Testing	Late, alag phase	Har sprint mein continuous
Documentation	Heavy	Light (working software pe focus)

🧠 **Yaad rakho:** Ek sawal sab decide karta hai: "Kya requirements badlengi?" Nahi → **Waterfall**. Haan / pata nahi → **Agile**.

Chapter 1 Exam Radar 🎯

- Analog vs digital (table + 2 graphs) — 3–4 marks
- Gate symbol + expression + truth table — 2–3 marks each
- Logic diagram of given Y + truth table — 4–5 marks
- K-map simplification

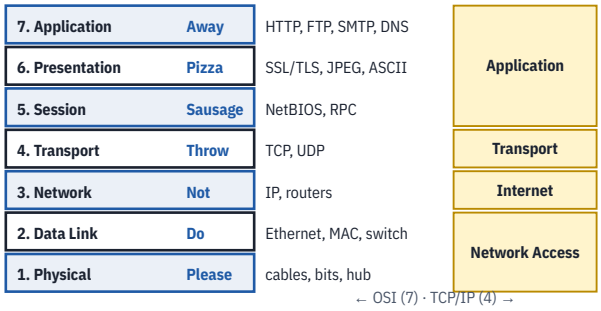
$$F = \sum m(\dots)$$

  — 5–6 marks
- SDLC 6 phases + deliverables — 5–8 marks
- Waterfall vs Agile (5-point table) / OSI 7 layers — long Q

1.3.1–1.3.4 Communication Models & OSI 7-Layer ★ Golden

Def (paper mein yehi likho): *A communication model explains how data travels from one device to another through a network.*

Data bhejna mushkil kaam hai, is liye **layers** mein baanta: har layer ek kaam kare aur agli ko de. **OSI** (Open Systems Interconnection) — **ISO** ne **1984** mein banaya, **7 layers**.



Layer	Kaam (function)
7 Application	User apps (browser, email) ko network services
6 Presentation	Translation, **encryption**, **compression**
5 Session	Session open, maintain, close
4 Transport	End-to-end reliable delivery, error check, ports. TCP = reliable, UDP = fast
3 Network	Logical (IP) addressing, routing — best path
2 Data Link	Frames, MAC address, error detection (switch)
1 Physical	Raw bits ko cable/wireless signals mein bhejna

🧠 **Yaad rakho:** Neeche se upar (1→7): "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way".

📝 **Paper mein:** Guaranteed long question: "Explain the 7 layers of OSI." Stack draw karo (7 upar), har layer ka 1 function + 1–2 protocols.

1.3.5–1.3.6 TCP/IP vs OSI

OSI = **reference** model (padhane/design ke liye). TCP/IP = **practical** model jo asal Internet pe chalta hai — sirf **4 layers**.

🧠 **Yaad rakho:** Ginti **3-1-1-2**: Application = 3 OSI layers, Transport = 1, Internet = 1, Network Access = 2. Total = 7 ✔

Checkpoint (40 min, closed book): (1) Sprint kya hai, kitna lamba? (2) Kis model mein customer sirf shuru/aakhir mein? (3) Encryption/compression kaunsi OSI layer? (4) TCP/IP mein kitni layers?

Jawab: 1. Chhota cycle jo kaam karta hissa deliver kare, 2–4 weeks · 2. Waterfall · 3. Presentation (6) · 4. Chaar. 60% se kam? Kal subah Golden repair — naya chapter nahi.

Day 06

Computational Thinking, Algorithm & Decomposition

CT · Algorithm · Algorithm vs Pseudocode · Decomposition · Pattern Recognition

★ **Must lock:** Algorithm, Algo vs Pseudocode, Decomposition

2.1 Computational Thinking (CT)

Def (paper mein yehi likho): *Computational Thinking is a systematic approach to solving problems.*

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

Pillar	Matlab (Urdish)
Decomposition	Bara masla chhote hisson mein todna
Pattern recognition	Jo cheez dohrai ja rahi ho / rule follow kare, pehchanna
Abstraction	Ghair-zaroori details chhupana / ignore karna
Algorithms	Saaf, step-by-step instructions banana

🧠 **Yaad rakho:** "**D**on't **P**ick **A**ny **A**pple" = Decomposition, Pattern, Abstraction, Algorithm.

2.2 Algorithm ★ Golden

Def (paper mein yehi likho): *An algorithm is a step-by-step logical procedure used to solve a problem.*

Khoobiyan: **Input** data se shuru · **Clear steps** har step ka ek hi matlab · **Definite end** limited steps ke baad khatam · **Output** result deta hai.

Ahmiyat (Importance)	Matlab
Structured approach	Mushkil kaam manageable ban jata hai
Optimisation	Time aur memory bachata hai (lakhon records search)
Reusability	Marks sort karne wala algorithm prices bhi sort kare
Accuracy & consistency	Same steps = hamesha same result, insani ghalti kam
Blueprint	Coding se pehle naqsha — jaise architect ka map

📝 **Paper mein:** "Define algorithm & write its importance" — definition word-for-word + **S-O-R-A-B** ke 5 points.

2.3.2 Pattern Recognition

Data mein regularity, repetition ya rule pehchanna. Teen sawal: **Kya dohra raha hai? Kya predictable tareeqe se badal raha hai? Kya ek rule sab pe lagega?**

Row 1: ★
Row 2: ★ ★
Row 3: ★ ★ ★
Row 4: ★ ★ ★ ★
Row 5: ★ ★ ★ ★ ★

Row N mein N stars → `for i in range(1,6): print("*"*i)`

🧠 **Yaad rakho:** "Same cheez bar bar, thore se farq ke saath" = **loop** socho.

2.2.2 Algorithm vs Pseudocode ★ Golden

Problem → Algorithm (**what**) → Pseudocode (**how**) → Program → Output

Point	Algorithm	Pseudocode
Definition	Step-by-step logical procedure to solve a problem	Human-readable representation of an algorithm using plain English & keywords
Nature	Conceptual, language-independent	Semi-formal, structured — **executable nahi**
Purpose	**Kya** karna hai (what)	**Kaise** implement hoga (how)
Form	Plain steps / numbered points	IF, FOR, WHILE jaise constructs
Audience	Koi bhi (non-programmer bhi)	Programmers / students
Format	Fixed format nahi, informal	Standard conventions follow

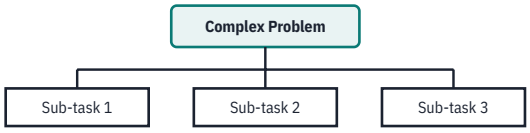
Algorithm:
1. Do numbers lo
2. Jama karo
3. Jawab dikhao

```
INPUT A, B
SET Sum ← A + B
PRINT Sum
```

🧠 **Yaad rakho:** Algorithm = recipe ("cake 350° pe bake karo"). Pseudocode = kitchen manual (SET, IF, PRINT). **Paper:** 4–5 marks, 5+ points ki table.

2.3.1 Decomposition ★ Golden

Def (paper mein yehi likho): *Decomposition is a problem-solving technique in which a complex task is broken down into smaller, manageable parts (sub-tasks).*



har sub-task alag solve → phir jodo = poora solution

Misaal — English seekhna: Vocabulary (I, eat, apple) → Grammar (Subject → Verb → Object) → Tenses (eat/ate) → Sentence ("I ate an apple.")

🧠 **Yaad rakho:** Bari roti ek nivale mein nahi khaa sakte — tukde karo. Decomposition = "roti todo" 🫓.

Drill: (1) CT ke 4 hisse? (2) Algorithm "language-independent" kyun? (3) FOR/WHILE keywords kaun use karta hai? (4) Decomposed hissa kya kehlata hai? (5) Pattern recognition se kaunsa concept?

Jawab: 1. Decomposition, pattern recognition, abstraction, algorithms · 2. Sirf logic batata hai, kisi language se bandha nahi · 3. Pseudocode · 4. Sub-task · 5. Loops | **5-mark:** Algorithm.

Abstraction, Bubble Sort & Selection Sort

Abstraction · Sorting & Searching · Bubble Sort · Selection Sort

★ **Must lock:** Bubble Sort, Selection Sort

2.3.3 Abstraction

Def (paper mein yehi likho): *Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem.*

Misaal — **chai banana**: Boil water → Add tea → Steep → Pour → Serve

1. Necessary (core logic)	2. Situation se badle (variables)	3. Noise (ignore)
Pani ubaalo, patti dalo, dam do, cup mein dalo, serve	Black/green/milk tea, cheeni/lemon, stove ya kettle	Kettle ka brand, cup ka rang, kitchen ya office

🧠 **Yaad rakho:** "Dhaancha rakho, sajawat hatao." Poochho: "Ye detail hata doon to bhi solution chalega?" Haan → noise.

2.4 Sorting vs Searching

Sorting	Searching
Items ko tarteeb mein lagana — ascending (0–9, A–Z) ya descending (9–0, Z–A)	List mein khaas element dhoondna — position ya maujoodgi (hai ya nahi?)
Bubble Sort, Selection Sort	Linear Search, Binary Search
Merit list: zyada se kam marks	Merit list mein apna roll no.

Swap (teesra variable chahiye): `temp = A → A = B → B = temp` (A=8, B=4 ⇒ A=4, B=8)

⚠ **Dhoka:** Textbook ki position 1 = Python ka index 0. Dono list same data type ki honi chahiye.

Complexity ki ginti (formulae)

N items ke liye Bubble/Selection: passes

$$= N - 1$$

; pass K mein comparisons

$$= N - K$$

.

$$\textbf{Total comparisons} = (N-1) + (N-2) + \cdots + 1 = \frac{N(N-1)}{2}$$

N = 5 → 4 passes,

$$\frac{5\times4}{2} = 10$$

comparisons. N = 6 → 5 passes, 15 comparisons.

Bubble vs Selection (exam table)

Point	Bubble Sort	Selection Sort
Compare	Parosi pairs (j, j+1)	Fixed DATA[K] vs baqi sab
Har pass ke baad	Sab se **bari** value **aakhir** mein	Sab se **chhoti** value **aage**
Inner loop	0 se N−1−i tak	outer + 1 se N tak
Stable?	Yes	No
Early stop	Swap na ho to ruk sakta hai → $O(N)$	Hamesha $O(N^2)$

🧠 **Yaad rakho: Stable** = barabar values ki asal tarteeb qaim rahe (do students ke 80 marks — pehle wala pehle hi rahe).

2.4.1 Bubble Sort ★ Golden

Def (paper mein yehi likho): *Bubble Sort repeatedly compares adjacent elements and swaps them if they are in the wrong order.*

Har pass mein sab se bari value "bulbula" ban ke aakhir tak pohanch jati hai. Pass 1 [8, 4, 1, 9, 3]:



Pass	List after
1	4, 1, 8, 3, **9**
2	1, 4, 3, **8, 9**
3	1, 3, **4, 8, 9**
4	**1, 3, 4, 8, 9** ✔

```
for i in range(n - 1):          # passes
    for j in range(n - 1 - i):
        if d[j] > d[j + 1]:
            d[j], d[j+1] = d[j+1], d[j]
```

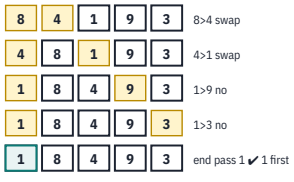
🧠 **Yaad rakho:** Bubble = **parosi** (j aur j+1). Descending ke liye sirf > ko < karo.

📝 **Paper mein:** "Sort … using Bubble Sort, show each pass" — har pass ki table: Pair · Compare · Action · List after.

2.4.1 Selection Sort ★ Golden

Def (paper mein yehi likho): *Selection Sort repeatedly selects the smallest element from the unsorted portion and moves it to its correct position at the front.*

Seats aage se bharo: position K fix, uske baad har element se compare, chhota mile to swap. Pass 1:



Pass	List after
1	**1**, 8, 4, 9, 3
2	**1, 3**, 8, 9, 4
3	**1, 3, 4**, 9, 8
4	**1, 3, 4, 8, 9** ✔

```
for k in range(n - 1):              # fixed position
    for p in range(k + 1, n):       # outer + 1 se
        if d[k] > d[p]:
            d[k], d[p] = d[p], d[k]
```

🧠 **Yaad rakho:** Selection: inner loop **outer + 1** se shuru, compare hamesha fixed **DATA[K]** se. Bubble: hamesha **parosi**. Selection sort **stable nahi**.

Drill: (1) Abstraction mein "noise"? (2) C, A, D, B descending? (3) 6 elements Bubble — passes & comparisons? (4) Selection mein inner loop kahan se?

Jawab: 1. Ghair-zaroori details · 2. D, C, B, A · 3. 5 passes; 5+4+3+2+1 = 15 · 4. outer + 1 (PTR = K + 1) | **5-mark:** Bubble Sort.

2.4.2 Linear Search (Sequential)

Def (paper mein yehi likho): *Linear Search examines each element in the list one by one, from the start, until the target is found or the list ends.*

Sorted aur unsorted dono pe chalta hai. **Flag** variable (ValueFound) 0 se shuru, mile to 1.

PTR = first → DATA[PTR] = Find? → Haan: flag=1, exit | Nahi: PTR+1 → List khatam? "Not Found"

Step	Element	== 7 ?	Result
1	8	No	aage
2	3	No	aage
3	6	No	aage
4	1	No	aage
5	7	Yes	**Found! exit**

```
data = [8, 3, 6, 1, 7]; find = 7; found = 0
for i in range(len(data)):
    if data[i] == find:
        found = 1
        break
print("Found" if found == 1 else "Not Found")
```

🧠 **Yaad rakho:** Queue mein dost dhoondna — aage se har chehra. Best case 1 check, worst case N checks.

2.5 Evaluation Criteria — C-E-C

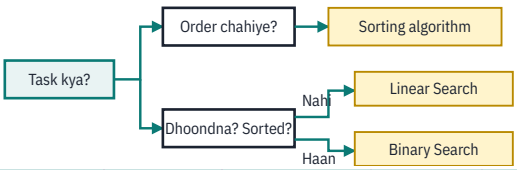
Criterion	Matlab
Correctness	Har input pe sahih jawab, aur khatam ho
Efficiency	Kam time + kam memory
Clarity	Parhne/samajhne mein aasaan

Even/odd check:

$$n \bmod 2 = 0 \Rightarrow$$

Even, warna Odd → `n % 2 == 0`. (3 → Odd, 6 → Even, 15 → Odd)

2.6 Kaunsa Algorithm?



Algorithm	Best	Worst	Sorted?	Stable?
Bubble	$O(N^2)$ / $O(N)$ *	$O(N^2)$	—	Yes
Selection	$O(N^2)$	$O(N^2)$	—	No
Linear	$O(1)$	$O(N)$	No	—
Binary	$O(1)$	$O(\log N)$	**Yes**	—

*early-stop check ke saath. Extra memory: sab ki "None".

🧠 **Yaad rakho:** Do keywords: "**sorted?**" aur "**kitna bara?**" Unsorted/chhota → Linear. Sorted + bara → Binary.

2.4.2 Binary Search ★ Golden

Def (paper mein yehi likho): *Binary Search is an efficient search algorithm that works by repeatedly dividing the search interval in half. (List must be sorted.)*

$$MID = \left\lfloor \frac{BEG + END}{2} \right\rfloor \quad (\text{INT} = \text{decimal hatao})$$

- Item = DATA[MID] → **found**, ruko
- Item < DATA[MID] → `END = MID - 1` (left half)
- Item > DATA[MID] → `BEG = MID + 1` (right half)
- Jab tak `BEG ≤ END` repeat; warna Not Found

DATA = 1…9, Item = **7**:



#	BEG	END	MID	D[MID]	Compare → Action
1	1	9	10/2 = 5	5	7 > 5 → BEG = 6
2	6	9	15/2 → 7	7	7 = 7 → **Found, LOC 7**

```
beg, end = 0, len(data) - 1
while beg <= end:
    mid = (beg + end) // 2
    if data[mid] == item:
        print("Found at", mid); break
    elif item < data[mid]:
        end = mid - 1
    else:
        beg = mid + 1
```

Har step pe aadhi list khatam: worst case

$$\approx \log_2 N$$

checks — 1000 items mein sirf ~10!

🧠 **Yaad rakho:** Dictionary mein lafz: beech se kholo, "pehle ya baad?", aadhi kitaab phenko. "**Pehle sorted, phir search.**"

📝 **Paper mein:** Sorted list + item diya jata hai: table BEG · END · MID · DATA[MID] · Compare · Action zaroor banao.

Checkpoint: (1) Linear Search ka doosra naam? (2) Binary Search ki zaroori shart? (3) Teen evaluation criteria? (4) Unsorted phone contacts — kaunsi search?

Jawab: 1. Sequential search · 2. List sorted honi chahiye · 3. Correctness, efficiency, clarity · 4. Linear Search | 60% se kam → kal Golden repair.

Day 09

Programming, Language Levels, IDE & VS Code

Programming · Language Types · Careers with Python · IDE · VS Code

★ **Must lock:** None — tables copy karo

3.1 Programming

Computer bohat tez hai magar "**dumb**" — khud kuch nahi soch sakta.

Term	Matlab
Program	A set of instructions given to a computer to perform a task
Programming language	Woh zubaan jis mein instructions likhte hain
Programming	Program likhne ka amal
Programmer	Jo program likhta hai

🧠 **Yaad rakho:** Program = recipe, programmer = recipe likhne wala, computer = cook jo har line bilkul waise hi follow kare. Recipe ghalat → khana ghalat.

3.1.1 Programming Language Types



Type	Description	Example
Low level	Computer ki apni zubaan (binary 0/1) ke qareeb; insaan ke liye bohat mushkil	Machine language
Mid level	Hardware access + **mnemonics** (MOV, ADD) — binary se aasaan	Assembly language
High level	English jaise alfaaz; parhna/likhna aasaan	C++, Java, Python

🧠 **Yaad rakho:** "**L**ow = **L**ike the machine, **H**igh = **H**uman." **M**id **M**iddle mein, **M**nemonics use kare.

3.2 Career Growth with Python

Field	Kaam
Software Development	Desktop & web applications
Data Science / Analysis	Data analyse, reports, predictions
AI & Machine Learning	Chatbots, recommendation ("videos you may like")
Web Development	Frameworks: **Django**, **Flask**
Automation	Repetitive kaam scripts se
Cybersecurity	Security tools, testing

🧠 **Yaad rakho:** Chhe careers = "**S-D-A-W-A-C**".

3.3 Integrated Development Environment (IDE)

Def (paper mein yehi likho): *An IDE is a software tool that helps programmers write, run and manage code in one place.*

Feature	Kaam
Editor	Code likhna aur edit karna
Syntax highlighting	Keywords, strings, numbers alag rangon mein
Auto-completion	Type karte hue suggestion (`pri` → `print`)
Run / Execute	Ek click mein program chalao
Debugger	Errors dhoondo, line pe ruko, variables dekho

🧠 **Yaad rakho:** IDE features = "**E-S-A-R-D**". "Integrated" = bohat se tools (editor, runner, debugger) **ek** program mein.

3.3.1 Visual Studio Code (VS Code)

Microsoft ka free, lightweight code editor. By default full IDE nahi — **extensions** (jaise Microsoft ki Python extension) lagao to IDE ban jata hai.

Install Python → Install VS Code → Extension: "Python" → New file hello.py → Code likho → ▶ Run



Element	Kaam
1. Top Menu Bar	Files, editing, navigation, workspace
2. Side Bar	Core tools: Explorer, Search, Source Control, Run & Debug, Extensions
3. Explorer	Project files/folders banao, kholo, manage karo
4. Search	File ya poore project mein find & replace
5. Source Control	**Git** se connect — changes track, undo, collaborate
6. Run & Debug	Code chalao, errors dhoondo, line pe pause
7. Extensions	VS Code ka "App Store" — Python, Prettier, Debugger

```
print("Hello, World!")     # pehla program
```

🧠 **Yaad rakho:** Extensions = phone ka App Store: naye apps = nayi salahiyat.

Drill: (1) Program kya hai? (2) `MOV AX, BX` kaunsa level? (3) Do Python web frameworks? (4) IDE mein "integrated" ka matlab? (5) VS Code ka kaunsa element Git se jurta hai?

Jawab: 1. Instructions ka set jo computer ko kaam karwaye · 2. Mid level (assembly) · 3. Django, Flask · 4. Saare tools ek program mein · 5. Source Control

Day 10

Python Basics, Variables, Data Types & Arithmetic

Fundamentals · Variables · Data Types · Input/Output · Arithmetic & Assignment

★ **Must lock:** Python fundamentals, Variables, Data Types, Arithmetic

3.4 Fundamentals of Python ★ Golden

Program data leta hai → statements se process → output. Python statements **upar se neeche**, ek ke baad ek (sequentially) chalata hai. **Statement** = ek mukammal instruction (`x = 5`).

```
# Comment: Python ignore karta hai
print("Hello, World!")
if 5 > 2:
    print("Five > two!")    # 4 spaces indent
```

- **Comment (#)**: code samjhane ke liye; run nahi hota.
- **Indentation** (4 spaces): block batata hai. C++/Java `{ }` use karti hain, Python spaces. Ghalat indent = **IndentationError**.
- Colon `:` ke baad naya block shuru.

🧠 **Yaad rakho:** Indentation = notebook ke points aur sub-points; sub-point thora right mein = "isi ka hissa".

3.4.1 Variables ★ Golden

Def (paper mein yehi likho): *A variable is a named space in computer memory used to store data.*

Rule	Valid ✔	Invalid ✘
Sirf letters, digits, underscore _	my_height	roll-no, my name, $pay
Letter ya _ se shuru, digit se nahi	_total, score1	2marks, 1name
Case-sensitive	`age` aur `Age` do alag variables	
Keyword nahi ho sakta	marks2026	class, if, for, True

🧠 **Yaad rakho:** Meaningful naam: `total_marks` > `tm`. Style = **snake_case**.

📝 **Paper mein:** "Which names are valid? Give reasons" — har naam ke saath toota hua rule likho ("2nd_place — digit se shuru").

3.4.2 Input & Output

```
name = input("Naam? ")          # hamesha str
age = int(input("Umar? "))      # number chahiye
print("Hi", name, age)          # comma = 1 space
print(f"{name} is {age} years")  # f-string
print("A", "B", sep="-")        # A-B
```

⚠ **Dhoka:** `input()` hamesha **string** deta hai. Maths ke liye `int(input())` / `float(input())` — andar se bahar parho.

3.4.1 Data Types in Python ★ Golden

Def (paper mein yehi likho): *A data type specifies the kind of data a variable holds.*

Name	Type	Description	Example
Integer	int	Poora number (+, −, 0)	`age = 16`
Float	float	Decimal point wala	`h = 5.7`
String	str	Text, `" "` ya `' '` mein	`year = "2026"`
Boolean	bool	True / False (capital T/F)	`ok = True`
List	list	Ordered, **changeable**, [ ]	`[85, 90, 75]`
Tuple	tuple	Ordered, **unchangeable**, ( )	`(10, 20)`
Dictionary	dict	key : value, { }	`{"age": 16}`
Set	set	Unordered, **unique** items	`{'a','e','i'}`

🧠 **Yaad rakho:** Brackets decide karte hain: **[ ]** List (badal sakti) · **( )** Tuple (taala laga 🔒) · **{k:v}** Dict · **{a, b}** Set (no duplicates). `type(x)` se check karo.

📝 **Paper mein:** "4 data types with examples" ya "List vs Tuple" (table: [ ] vs ( ), changeable vs unchangeable, example).

3.4.3 Arithmetic & Assignment Operators ★ Golden

Operand = jis value pe kaam ho; **operator** = symbol. `10 + 20`: 10, 20 operands, + operator → expression = 30.

Op	Kaam	Example	Result
+ − *	Jama, tafreeq, zarb	5+3, 10−4, 4*2	8, 6, 8
/	Division (hamesha **float**)	9 / 2 ; 8 / 2	4.5 ; 4.0
//	Floor division (decimal gira do)	9 // 2	4
%	Modulus (baqi / remainder)	9 % 2	1
**	Power	2 ** 3	8

$$17 = 5 \times 3 + 2 \Rightarrow$$

`17 // 5 = 3`, `17 % 5 = 2`. Even: `n % 2 == 0`.

Assignment	Matlab	x = 10 ke baad
x += 3	x = x + 3	13
x -= 3	x = x − 3	7
x *= 2	x = x * 2	20
x //= 3 / x %= 3	floor / baqi	3 / 1
x **= 2	x = x ** 2	100

Precedence: `()` → `**` → `*` `/` `//` `%` → `+` `−`. `2 + 3 * 2 ** 2`

$$= 2 + 3 \times 4 = 14$$

🧠 **Yaad rakho:** Teen division: **/** = poora jawab (float), **//** = sirf quotient, **%** = sirf baqi.

Drill: Valid/invalid: `_total, roll-no, class, marks2026, 2marks` · Types: `3.0, "3", 3, False, [3]` · `print("3"*4, 3*4)`?

Jawab: _total ✔, roll-no ✘ (hyphen), class ✘ (keyword), marks2026 ✔, 2marks ✘ (digit se shuru) · float, str, int, bool, list · `3333  12` (string × number = repeat) | **5-mark:** Variables / Data types.

Day 11

Relational, Logical, Bitwise & if / if-else

Relational · Logical · Membership · Bitwise · Control Structures · Sequence · if/if-else

★ **Must lock:** Operators, Control Structures, if & if-else

3.4.3 Relational, Logical, Membership ★ Golden

Relational operators do values compare karke **True/False** dete hain (a = 10, b = 5):

Op	Matlab	Example	Result
==	barabar	a == b	False
!=	barabar nahi	a != b	True
>	bara	a > b	True
<	chhota	a < b	False
>=	bara ya barabar	a >= 10	True
<=	chhota ya barabar	a <= b	False

A	B	A and B	A or B	not A
True	True	True	True	False
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True

Worked (a=10, b=5): `a > b and b > 3` → True and True → **True** · `not a == 10 or b < 2` → not True or False → False or False → **False**.

Membership: `"a" in "apple"` → True · `5 not in [1,2,3]` → True. Order: **not → and → or**.

🧠 **Yaad rakho:** **and** = sakht teacher (homework *aur* attendance dono). **or** = dukaan (cash *ya* card).

⚠ **Dhoka:** = value **rakhta** hai, == **compare** karta hai.

3.5 Control Structures ★ Golden

Def (paper mein yehi likho): *Control structures decide the order in which statements are executed (flow of control).*

Type	Kaam	Python
Sequence	Upar se neeche, ek ke baad ek	default
Selection	Condition pe raasta chuno	if, else, elif
Repetition	Steps dohrao	for, while

🧠 **Yaad rakho:** "**S-S-R**" = Step, Select, Repeat.

3.5.1 Sequence

```
a = int(input("First: "))    # Step 1: Input
b = int(input("Second: "))   # Step 2: Input
total = a + b                # Step 3: Process
print("Sum =", total)        # Step 4: Output
```

Koi statement skip ya repeat nahi. Pattern: **Input → Process → Output**.

3.4.3 Bitwise Operators ★ Golden

Numbers ko **binary bits** bana ke column-by-column kaam:

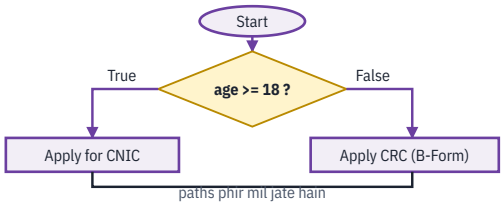
Op	Rule	12 (1100) op 10 (1010)	=
& AND	1 sirf jab dono 1	1000	**8**
| OR	1 jab koi ek 1	1110	**14**
^ XOR	1 jab bits alag	0110	**6**
~ NOT	sab bits ulti	$\sim x = -(x+1)$: ~5 = **−6**, ~10 = −11	
<< left	$x \ll n = x \times 2^n$	3 << 3 = 24	
>> right	$x \gg n = \lfloor x/2^n \rfloor$	7 >> 1 = 3 (111 → 11)	

```
  1 1 0 0   (12)
& 1 0 1 0   (10)
---------
  1 0 0 0   = 8
```

📝 **Paper mein:** "Evaluate 12 & 10, 12 | 10, 12 ^ 10, ~5, 7 >> 1 showing binary working" — marks binary conversion aur column working ke hain, sirf jawab ke nahi. Bits barabar (4-bit) likho.

3.5.2 Selection: if & if-else ★ Golden

Condition (True/False) pe program ka raasta badalta hai. **if**: True ho to block chale, False ho to bilkul skip.




```
n = int(input("Number: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

🧠 **Yaad rakho:** `if` aur `else` ke baad **colon :** aur andar ka code **indent**. `else` ki apni condition **kabhi nahi** hoti.

📝 **Paper mein:** "Program: even/odd, positive/negative, pass/fail" — `int(input())`, condition, dono branches poore likho.

Drill: (1) `print(10 > 5 and 5 > 3)`? (2) `12 & 10, 12 | 10, 12 ^ 10, ~5`? (3) Teen control structures? (4) `x=4; y=x*2; x=10; print(x, y)`?

Jawab: 1. True · 2. 8, 14, 6, −6 · 3. Sequence, selection, repetition · 4. `10 8` — y tab bana jab x = 4 tha; baad mein x badalne se y nahi badalta.

Day 12

if-elif-else, for/range, while, break/continue & Nested Loops

Selection ladder · Nested if · for & range() · while · break/continue · Nested loops · Shopping Bill

★ **Must lock:** if-elif-else, for, while, nested loops

3.5.2 if-elif-else & Nested if ★ Golden

elif = "else if". Python upar se neeche check karta hai; **pehli True** condition ka block chala ke baqi sab ignore. Koi True na ho → else.

```
marks = int(input("Marks: "))
if marks >= 80:
    print("A-1 Grade")
elif marks >= 70:
    print("A Grade")
elif marks >= 60:
    print("B Grade")
else:
    print("Fail or below B")
```

marks	>=80	>=70	>=60	Output
92	True	—	—	A-1
75	False	True	—	A
60	False	False	True	B
41	False	False	False	Fail

Nested if = if ke andar if (largest of three):

```
if a > b:
    if a > c: print("a largest")
    else: print("c largest")
```

🧠 **Yaad rakho:** Seerhi (ladder): sab se **sakht** condition pehle (zyada marks pehle). Pehli fit hone wali seerhi pe utar jao.

3.5.3 for Loop & range() ★ Golden

Loop = block bar bar chale; har ek chakkar = **iteration**. **for** tab jab iterations ki ginti **maloom** ho.

Syntax	Kaam	Example	Values
range(stop)	0 se stop tak (stop shamil nahi)	range(3)	0 1 2
range(start, stop)	start se stop−1	range(2, 5)	2 3 4
range(start, stop, step)	step ki chhalang	range(1, 6, 2)	1 3 5
Ulta (negative step)		range(10, 0, −3)	10 7 4 1

Values ki ginti:

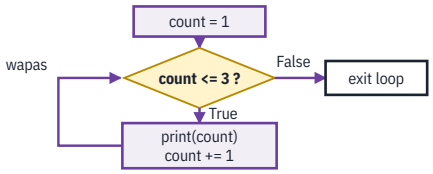
$$\text{range}(a, b) \Rightarrow b - a$$

values. Sum 1..n: `for i in range(1, n+1): s += i`

🧠 **Yaad rakho:** range **ek pehle ruk** jata hai: range(1, 11) = 1...10. n shamil karna ho to `n + 1`.

3.5.3 while, break, continue ★ Golden

while = jab tak condition True rahe, block dohrao.



count	count<=3?	Printed	After update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends

```
for i in range(1, 6):
    if i == 3:
        break
    print(i)
# Output: 1 2
```

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
# Output: 1 2 4 5
```

🧠 **Yaad rakho:** **break** = "kamre se bahar" (loop khatam). **continue** = "ye baari chhoro" (agli iteration). Har while ke liye **I-T-U**: Initialise, Test, Update.

⚠ **Dhoka:** Update bhool gaye (`count += 1`) → **infinite loop** — kabhi khatam nahi hoga.

3.5.3 Nested Loops & for vs while ★ Golden

Loop ke andar loop: **inner** loop outer ki **har** iteration pe poora chalta hai. Runs

$$= \text{outer} \times \text{inner}$$

→ 2 × 3 = 6.

```
for outer in range(1, 3):       # 2 times
    for inner in range(1, 4):   # 3 times each
        print(f"Outer: {outer}, Inner: {inner}")
```

for loop	while loop
Iterations ki ginti **maloom**	Ginti **na-maloom** (condition-based)
Variable khud update	Manual update (infinite loop ka khatra)
Counting ke liye chhota, saaf	Dynamic: "jab tak user Stop na kahe"

🧠 **Yaad rakho:** "Kitni dafa maloom?" Haan → **for**. "Sirf rukna maloom?" → **while**. Nested = ghari: minute ki sui (inner) poora chakkar, ghante ki sui (outer) ek qadam.

Practical — Shopping Bill: while loop + flag `shopping = True`; `item.lower() == "done"` → flag False; price <= 0 → invalid, add nahi; total > 5000 → discount. Trace: Rice 3000 → total 3000 · Soap −50 ✘ · Oil 2500 → 5500 · DONE → stop.

Drill jawab: `temp=30` ladder (>35, >25, >15) → **Warm** · `range(10,0,−3)` → 10 7 4 1 · n: 1→3→9→27 (27 < 20 False) → **27** · 3 × 2 nested → **6** · ek item 4000 + done → Final 4000.0 (5000 se kam, discount nahi).

Libraries, Built-in & Third-Party, Debugging

Libraries · math / random / datetime · pip packages · Syntax, Runtime & Logical errors

★ **Must lock:** None — tables copy karo

3.6 Libraries in Python

Def (paper mein yehi likho): *A library (module) is a collection of pre-written code that offers ready-made functions.*

Square root khud likhne ki bajaye `import math` → `math.sqrt()`. Faide: time bachta, bugs kam, doosron ke kaam pe khare ho.

Tareeqa	Syntax	Call kaise
1. Poora module	`import math`	`math.sqrt(25)`
2. Alias (nickname)	`import math as m`	`m.sqrt(16)`
3. Ek function	`from math import sqrt`	`sqrt(36)`
4. Kuch functions	`from math import sqrt, pi`	`sqrt(9), pi`
5. Sab kuch (*)	`from math import *`	`sqrt(64)` — avoid

⚠ **Dhoka:** `from math import *` har naam program mein dump kar deta hai — naam takra sakte hain.

3.6.1 Built-in Libraries

Python ke saath pehle se install — sirf **import**, **pip nahi**. Teen zaroori: **math, random, datetime**.

Function	Kaam	Result
`math.pi`	π	3.141592...
`math.ceil(4.3)`	Upar round ($\lceil x \rceil$)	5
`math.floor(4.9)`	Neeche round ($\lfloor x \rfloor$)	4
`math.pow(2, 4)`	2^4 (float)	16.0
`math.sqrt(81)`	$\sqrt{81}$	9.0
`random.randint(1, 6)`	1–6 random (dono shamil)	jaise 4
`random.choice(list)`	List se koi ek	random item
`datetime.date.today()`	Aaj ki date	2026-10-12

🧠 **Yaad rakho:** **ceil** = chhat (upar), **floor** = farsh (neeche). `round()` built-in hai (math mein nahi) — qareeb tareen integer.

3.6.2 Third-Party Libraries

Community ne likhi; Python ke saath nahi aati. Terminal mein `pip install library_name`, phir import.

Library	Primary use
NumPy	Heavy numerical computing, bare arrays
Pandas	Data analysis; table jaisa data (CSV/Excel)
Matplotlib	Data visualisation — graphs, charts
TensorFlow / PyTorch	AI aur machine learning

```
pip install pandas        # terminal mein
import pandas as pd       # Python file mein
```

⚠ **Dhoka:** `pip` terminal command hai, Python statement nahi — .py file mein mat likho.

3.7 Debugging & Error Types

Def (paper mein yehi likho): *Debugging is the process of identifying, tracing and fixing bugs so a program works correctly.*

Type	Kab aata hai	Nishani	Example
Syntax	Program shuru hone se **pehle**	Python chalne se inkaar	`print("Hello"` — missing
Runtime (exception)	Chalte **hue**	Crash + traceback	`10 / 0` → ZeroDivisionError; `int("abc")` → ValueError
Logical	Program "kamyabi" se khatam	Output **ghalat**	`area = l + w` (* hona chahiye)

1. Syntax errors theek → 2. Run karo (runtime) →
3. Haath se hal kiye example se output milao (logical)

📝 **Paper mein:** "Identify the error type": (1) Shuru hi nahi hota? → **syntax**. (2) Beech mein crash? → **runtime**. (3) Chalta hai par jawab ghalat? → **logical**.

Chapter 3 Exam Radar 🎯

- Valid/invalid variable names + reasons
- Data types (4 with examples), list vs tuple
- Evaluate: `17 // 5, 17 % 5, 2 ** 4`, bitwise with binary working
- Programs: even/odd, grade ladder, largest of 3, table, sum 1..n
- break vs continue; for vs while; predict output of range()
- Identify error type

Checkpoint: (1) Woh line jis se `ceil` bina `math.` ke call ho? (2) `math.ceil(4.0)`? (3) Class marks ka bar chart — kaunsi library? (4) `print("Hi"` — kaunsa error?

Jawab: 1. `from math import ceil` · 2. 4 · 3. Matplotlib · 4. Syntax error | 60% se kam → kal Golden repair.

Day 14

Data, DBMS, Tables & Keys

Data vs Information · DBMS · Tables, Records, Fields · Keys & Integrity · RDBMS

★ **Must lock:** Keys and Integrity Constraints

4.1.1 Data, Information, Database

Data = unprocessed facts, figures, observations. **Quantitative** (numbers: 80, 75, 90) ya **qualitative** (descriptions: cars ke rang).

Data → Processing → Information → Decision

Aspect	Data	Information
Definition	Raw facts & figures	Processed, meaningful data
Example	85, 90, 75	Average = $\frac{85+90+75}{3} = 83.3\%$
Faida	Akele mein kam	Decisions ke liye useful

Database = organised, related data ka collection.

🧠 **Yaad rakho:** Exam line: "Data is raw; information is processed data used for decision-making." Saath mein ek example pair zaroor.

4.1.2 DBMS

Def (paper mein yehi likho): *A DBMS is software that lets you create, manage and interact with databases.*

Misaalein: **MS Access, MySQL, Oracle, SQLite**.

Faida	Matlab / misaal
Less redundancy	Department ka naam ek dafa, har student row mein nahi
Consistency	Phone no. ek jagah badlo → har screen pe naya
Security	Admin marks dekhe, office staff sirf naam
Integrity	Rules: duplicate roll no. ya khali PK nahi
Sharing	Bohat se users ek saath

4.2 Tables, Records, Fields

Table / Relation: STUDENT

Roll No (PK)	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

Record → (row/tuple)

↑ Field (column / attribute)

3 records × 4 fields

Term	Doosre naam	Matlab
Table	Relation	Ek qisam ki cheez ke facts (Students)
Record	Row, tuple	Ek mukammal instance — Ali ki sab info
Field	Column, attribute	Ek qisam ka fact — poora Name column

🧠 **Yaad rakho:** row = record = **ek banda**. column = field = admission form ka **ek sawal**.

4.3.1 Keys & Integrity Constraints ★ Golden

Def (paper mein yehi likho): *A key is a field (or set of fields) that uniquely identifies records and links tables. Integrity constraints are rules that keep keys honest.*

Key	Matlab	Misaal
Candidate key	Har woh field jo row ko unique pehchan sake; table mein kai ho sakti hain	RollNo, BFormNo
Primary key (PK)	Chuni hui candidate key — **unique + NOT NULL**; table mein sirf **ek**	Student.RollNo
Alternate key	Candidate key jo PK nahi bani	Student.BFormNo
Foreign key (FK)	Doosri table ki PK ko refer kare; tables jore	Result.RollNo → Student
Composite key	Do ya zyada fields mil ke PK	ExamID + RollNo

Primary key	Foreign key
Apni table mein row ko pehchane	Doosri table se link kare
Unique — duplicate nahi	Duplicate ho sakti hai
NULL nahi ho sakti	NULL ho sakti hai (agar allowed)
Table mein sirf ek	Kai ho sakti hain

Constraints: **Entity integrity** PK unique & not null · **Referential integrity** FK ki value parent mein maujood · **Domain** sahih data type/range.

⚠ **Dhoka:** Phone number **bura PK** hai: log number badalte, do bhai-behan ek share karein, kuch ke paas hota hi nahi (NULL).

📝 **Paper mein:** "Differentiate primary & foreign key" (4-point table) aur "PK NULL kyun nahi ho sakti?" → NULL ka matlab "is row ki koi pehchan nahi".

4.3 RDBMS

Def (paper mein yehi likho): *An RDBMS stores data in several related tables and uses primary and foreign keys to keep relationships.*

- Data ko related tables mein baant ke **redundancy kam**
- Ek waqt mein bohat users, permissions ke saath
- **SQL** (Structured Query Language) se query: `SELECT, INSERT, UPDATE, DELETE`

```
SELECT Name, Grade FROM Student
WHERE Course = &#x27;CS';
```

Drill: (1) "Nadia, Class 11, 72 marks" — data ya information? (2) Do DBMS programs? (3) Upar wali table mein records/fields? (4) Kya table ki do PK ho sakti hain? (5) RDBMS ki query language?

Jawab: 1. Data (ek student ke raw facts) · 2. MS Access, MySQL · 3. 3 records, 4 fields · 4. Nahi — sirf ek (composite ho sakti hai) · 5. SQL | **5-mark:** Keys & Integrity Constraints.

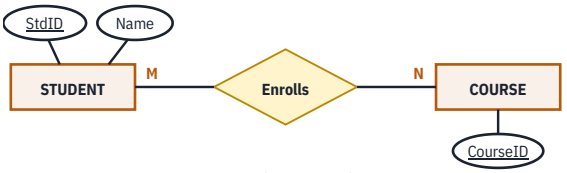
ER Model, Referential Integrity & Schema

ER Model · Referential Integrity · Relational Schema · Library Case Study · Database Objects

★ **Must lock:** ER Model, Referential Integrity

4.4 Entity Relationship Model (ER) ★ Golden

Def (paper mein yehi likho): *The ER model is a blueprint that shows what data will be stored (entities, attributes) and how it connects (relationships) before tables are created.*



▭ entity · ◇ relationship · ⬭ attribute (PK underline)

Component	Matlab	Symbol	Misaal
Entity	Real-world cheez jiska data rakhein	Rectangle ▭	STUDENT, COURSE
Attribute	Entity ki property; PK underline	Oval ⬭	StudentID, Name
Relationship	Do entities ka taalluq	Diamond ◇	Enrolls, Teaches

Cardinality	Matlab	Misaal
1 : 1	Ek ka sirf ek	Student — B-Form
1 : M	Ek ke bohat	Teacher — Classes
M : N	Bohat ke bohat	Students — Courses

⚠ **Dhoka: M:N rule:** M:N seedha table mein nahi banta — beech mein **junction table** banao jo do **1:M** links bana de.

📝 **Paper mein:** Line ke dono siron pe cardinality (1, M) likho. M:N ho to "resolve kaise?" — junction entity draw karo jis mein dono FK.

4.6 Relational Schema

Database ka formal likha hua design: table naam, fields, PK, FK, relationships. Access kholne se **pehle** likho.

```
Department (DeptID PK, DeptName)
Student (RollNo PK, Name, Class,
         DeptID FK → Department)
Course (CourseID PK, Title, Credit,
        DeptID FK → Department)
```

🧠 **Yaad rakho:** Schema chhorna = naqshe ke baghair ghar banana — tooti phooti structure, theek karna dard-e-sar.

4.5 Referential Integrity ★ Golden

Def (paper mein yehi likho): *Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent table (or be NULL if allowed).*

Parent: Student

StudentID (PK)	Name
101	Ali
102	Sara

Child: Result

StudentID (FK)	Marks
101 ✔	78
102 ✔	85
999 ✘ orphan	60

999 parent mein nahi → Access insert hi nahi karne dega

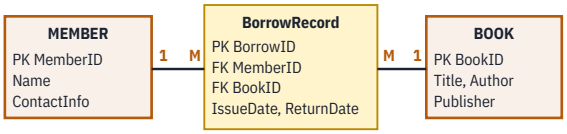
- Result mein student 999 nahi likh sakte agar Student mein 999 nahi.
- Student 101 delete nahi kar sakte jab tak uske results maujood (pehle child rows handle karo).
- **Orphan record** = child row jiski FK value parent mein nahi.

Option	Kya karta hai
Cascade Update	Parent ka PK badla (101 → 111) → child ki FK khud badal jaye
Cascade Delete	Parent row delete → us ke sab child rows bhi delete

⚠ **Dhoka:** Cascade Delete taqatwar aur **khatarnak** hai — ek student delete, saare results ghaib.

4.7 Case Study — Library System

Book: ID, title, author, publisher. Member (student/teacher): ID, name, contact. Member bohat books le, book bohat members le → **M:N** → BorrowRecord junction.



BorrowRecord: 5001 · 102 · B-17 · 2026-10-05 · (ReturnDate)

4.8 Database Objects (Access)

Object	Kaam
Table	Data store — buniyaad, sab is pe depend
Form	Data enter/edit/view ki friendly screen
Query	Sawal poochho: filter, sort, calculate ("80% se upar")
Report	Print/share ke liye formatted output — read only

🧠 **Yaad rakho:** Store → Enter → Search → Print = Tables, Forms, Queries, Reports.

Drill: (1) Relationship ka symbol? (2) Orphan record? (3) Schema mein do cheezein? (4) BorrowRecord kyun? (5) Printed fee slip — kaunsa object?

Jawab: 1. Diamond · 2. Child row jiski FK parent mein nahi · 3. Tables/fields aur keys/relationships · 4. M:N ko do 1:M mein todne ke liye · 5. Report | **5-mark:** ER Model / Referential Integrity.

MS Access: Tables, Forms, Queries & Charts

Tables in Access · Forms · Queries · Summarisation · Charts

★ **Must lock:** Creating Queries in MS Access

4.9 Tables in MS Access

Pehle plan: maqsad, fields ki list, har field ka data type, PK kaun.

Create tab → Table → Field1, Field2... double-click rename → Data type set → PK 🔑

Data type	Kis ke liye
Short Text	Naam, address, **phone no.** (255 characters tak)
Number	Jin pe hisaab ho: Age, Marks, Quantity
Date/Time	DOB, admission date
Currency	Fees, salary
AutoNumber	Access khud unique sequential number — behtareen PK
Yes/No	Haan/nahi (Fee paid?)

⚠ **Dhoka:** Phone number dikhta number hai par **Short Text** hai — us ko jama nahi karte, aur shuru ka 0 gir jata hai.

4.10 Forms

Create → Form Wizard → Table chuno → Fields: > (ek) / >> (sab) → Layout

Columnar = ek waqt mein ek record — data entry ke liye best. Form ko **data source** (table/query) chahiye.

4.11.4 Summarisation (Totals Σ)

Grouping = same value wali rows ikatthi (sab CS students); phir **aggregate** group ka number nikale. Access mein **Totals (Σ)** → Total row.

Function	Hisaab
COUNT()	Kitne records (IT mein students)
SUM()	Number field ka total (fees)
AVG()	Mean $\bar{x} = \frac{\sum x}{n}$
MAX() / MIN()	Group mein sab se zyada / kam

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

4.11 Creating Queries in MS Access ★ Golden

Def (paper mein yehi likho): *A query retrieves, filters and analyses data without changing the original tables.*

Create → Query Design → Show Table: Student → Fields double-click → Criteria → Run (!)

Field:	Name	Department	Marks
Table:	Student	Student	Student
Show:	☑	☑	☑
Criteria:	"Ali"	"Computer Science"	>=80
or:		"IT"	

Gold row = AND (same row): Name "Ali" **aur** Dept "Computer Science" aur Marks ≥ 80 · "IT" or: row mein = OR

Goal	Kahan likho	Misaal
Single criterion	Criteria row, ek column	Department: "Computer Science"
AND (dono match)	**Same** Criteria row, do columns	Name "Ali" + Dept "Computer Science"
OR (koi bhi)	Criteria row **aur** or: row	"IT" or "Computer Science"
Range	Criteria	`>=80`, `Between 50 And 80`
Wildcards	Like	`Like "A*"` (A se shuru), ? = ek harf

⚠ **Dhoka:** Text criteria ko quotes chahiye: `"IT"`. Numbers ko nahi: `>=80`.

📝 **Paper mein:** AND = grid ki **same row**; OR = **alag rows**. Examiner khali grid de ke Criteria aur or: bharwata hai.

4.11.6 Charts in MS Access



Chart	Best for
Column (vertical)	Side-by-side comparison (2024/2025/2026 admissions)
Bar (horizontal)	Lambe naam wali categories (English, Mathematics...)
Pie	Ek whole ke hisse (CS vs IT vs Eng %) — kam slices
Line	Waqt ke saath trend (har mahine visitors)

📝 **Paper mein:** "Which chart?" — line = time, pie = composition, bar/column = categories compare. Ek wajah saath likho.

Checkpoint: (1) StudentID Access khud banaye — data type? (2) Form bina table/query ke? (3) Marks kam az kam 80 — criteria? (4) Department ke neeche Total setting? (5) Har mahine website visitors — chart?

Jawab: 1. AutoNumber · 2. Nahi — data source chahiye · 3. >=80 Marks ke neeche · 4. Group By · 5. Line | 60% se kam → kal Golden repair.

Day 17

AI, IoT & Data Analytics

Computing Impacts · AI · IoT & Components · Data Analytics · IoT vs AI vs Analytics

★ **Must lock:** AI, IoT & Components

5.1 Computing Impacts — Introduction

Computing = computers aur computer-based devices se information store, process aur share karna.

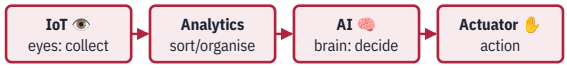
Schools: online admissions, e-books, board results · **Hospitals**: patient records, digital X-rays · **Banks**: ATMs, **Raast**, JazzCash, Easypaisa · **Farms**: weather apps, soil sensors · **Industries**: automation.

5.1.1 Artificial Intelligence (AI) ★ Golden

Def (paper mein yehi likho): *Artificial Intelligence is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.*

Artificial = insaan ne banaya, qudrati nahi. **Intelligence** = sochne wale kaam kar sake.

Misaalein: phone ka face unlock · YouTube recommendations · Safe City number-plate recognition (e-challan) · chatbots.



🧠 **Yaad rakho:** "Eyes – Sort – Brain – Do." Cricket fielder: aankh ball dekhe (IoT), dimagh hisaab lagaye (analytics + AI), haath pakre (actuator).

📝 **Paper mein:** "Define AI + 2 examples" (2–3 marks) · "AI, IoT, Data Analytics mil ke kaise kaam karte hain?" (4–5 marks) — chain diagram + har ek ka role.

5.1.3 Data Analytics & Components

Def (paper mein yehi likho): *Data Analytics is the process of collecting, organising, cleaning, studying and presenting data to find useful information.*



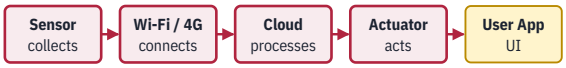
Cleaning = duplicate rows hatao (Hina 2 dafa), ghalat values theek (Kashif 250/100), missing values sambhalo (Sana —).

🧠 **Yaad rakho:** "**C**ows **S**it **C**almly **A**nd **V**ote".

5.1.2 Internet of Things (IoT) & Components ★ Golden

Def (paper mein yehi likho): *The Internet of Things is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.*

"Things" = aam cheezein (ghari, camera, car, pankha, water pump) jinhein **sensors + internet** mil gaya. Smartwatch dil ki dharkan le ke phone pe bheje.



User app se commands wapas cloud ko bhi ja sakti hain · data processing cloud ke andar

#	Component	Kaam	Misaal
1	**Sensors**	Physical duniya se data — system ki "aankh aur kaan"	Temperature, motion, fingerprint, smoke, soil moisture
2	**Actuators**	Digital signal → physical action	Motor, switch, auto door, water valve, fan controller
3	**Connectivity**	Data bhejna/lena; iske baghair sensor baat nahi kar sakta	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	**Data Processing**	Raw data → info: collect, organise, analyse, decide	Kamra garam? → AC on
5	**Cloud Computing**	Internet pe storage/services — IoT ka bhaari data	Google Drive, iCloud, LMS
6	**User Interface**	User info dekhe aur command de	Mobile app, touch screen, voice

🧠 **Yaad rakho:** **Sensor senses** (input), **Actuator acts** (output) — lafz hi jawab hain.

📝 **Paper mein:** "Define IoT & explain components" — sab se frequent long question. Definition ke 3 ideas (physical devices, internet se connected, khud-ba-khud kaam) + 6 components + diagram.

IoT vs AI vs Data Analytics

Aspect	IoT	AI	Data Analytics
Kya hai / kaam	Internet se jure physical devices — **hardware**	Machines sochein/decide karein — smart **software**	Data jaanch ke patterns — **data** ke saath
Maqsad → Output	Collection, automation → **raw readings**	Decisions, predictions → **decision/action**	Insights → **reports, graphs**
Role · Pakistan	Aankh & haath · Safe City cameras	Dimagh · number plate → e-challan	Organiser · kis sarak pe zyada violations

Drill: (1) Pakistan mein 4 jagah jahan computing zaroori? (2) AI "artificial" kyun? (3) Sensor ya actuator: fingerprint scanner, auto door, motion detector, motor? (4) Data analytics ke 5 components tarteeb se? (5) Kaun hardware ke saath kaam karta hai?

Jawab: 1. Schools, hospitals, banks, farms (industries) · 2. Insaan ne banaya · 3. sensor, actuator, sensor, actuator · 4. Collection, Storage, Cleaning, Analysis, Visualization · 5. IoT | **5-mark:** IoT & components.

IoT Uses, AI in Education & Information Sources

Uses of IoT · AI in Education (Pakistan) · Careers · Primary, Secondary, Tertiary Sources

★ **Must lock:** Information Sources

5.2 Uses of IoT in Different Areas

IoT **efficiency** barhata (kam mehnat, zyada kaam), waqt bachata, zindagi behtar karta; mulk ke **infrastructure** (sarak, bijli, pani, hospitals) mein ahem.

Soil sensor → Cloud AI: pani chahiye? → Water pump ON

Area	Misaal
Cities (smart)	Smart traffic signals, Safe City cameras, **smart bins** (bhar jayein to alert)
Health	Smartwatch heart-rate, remote patient monitoring
Industry	Machines ki monitoring, kharabi se pehle alert
Education	Smart classroom, automatic attendance
Farming	Soil moisture sensor + auto irrigation
Homes	Smart lights, AC, door locks — phone se

🧠 **Yaad rakho:** Chhe areas = "**CHIEF H**": Cities, Health, Industry, Education, Farming, Homes.

5.2.1 AI in Education in Pakistan

Teaching, learning aur **administration** (admissions, records, timetables) behtar.

Practical uses	Benefits	Challenges
LMS (Google Classroom, Moodle)	Apni speed se seekhna	Rural internet kam
Automated grading & digital testing	Madad ke mohtaaj students pehchanna	Computers/devices ki kami
AI career counselling	Disabled students ki madad	Teacher training chahiye
—	—	Data privacy & security

⚠ **Dhoka:** "AI teachers ki jagah le lega" likhna **ghalat** — AI teacher ki **madad** karta hai.

5.2.2 Careers in IoT & AI

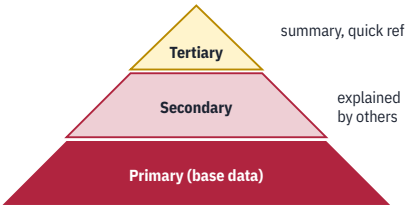
Employers: software houses, telecom, banks, startups. **IoT Engineer** = smart systems/connected devices design (smart irrigation kit). **Data Analyst** = IoT data se reports. Aur: AI/ML engineer, network engineer.

🧠 **Yaad rakho:** Skills = "**PNEP**": Programming (Python, C++), Networking (Wi-Fi, protocols), Electronics, Problem-solving. Pehli skill isi saal Python se!

5.3 Information Sources ★ Golden

Def (paper mein yehi likho): *Information sources are ways to find knowledge and data. They are primary (firsthand), secondary (explained by others) and tertiary (summarised).*

Internet pe bohat info hai — magar **sab sahih nahi**.



Primary buniyaad hai; har upar wala level neeche walon se bana aur zyada summarised

Primary (firsthand)	Secondary (doosron ki wazahat)	Tertiary (khulasa)
Dadu ke flood victim ka interview	2022 floods pe akhbar ki report	Encyclopedia: "Floods in Pakistan"
Event ki photograph	Event samjhane wala textbook chapter	Wikipedia article
Diary, letter	Historian ki likhi biography	Dictionary
NADRA records (CNIC)	PBS census analysis report	Handbook / almanac
Asal survey responses	Survey pe research paper	Research papers ki index
IoT weather station readings	TV news weather analysis	School climate fact sheet
Quaid-e-Azam ki asal taqreer recording	Taqreer pe article	Encyclopedia mein unki zindagi

🧠 **Yaad rakho:** **Author – Date – References:** online article mein author ka naam, publication date ya references na hon → bharosa mat karo.

📝 **Paper mein:** "Differentiate primary, secondary, tertiary + 2 examples each" (long/short) aur "Classify these sources" (MCQ). Pyramid draw karo.

5-mark frame: IoT

Define	IoT is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.
Explain	Sensors data lete, connectivity bhejti, cloud process karta, actuator action leta, app se user control.
Example	Smart irrigation: soil dry → pump ON; smartwatch heart-rate.
Diagram	Sensor → Wi-Fi → Cloud → Actuator → App chain.
Working	6 components ki table + ek field (CHIEF H) ka use.

Drill: (1) Smart bins waste management mein kaise madad? (2) AI in education ke 3 practical uses? (3) IoT Engineer kya karta hai? (4) Match ke baad cricketer ka video interview — P, S ya T?

Jawab: 1. Bharne pe alert — sirf bhari bins khali hoti hain · 2. LMS, automated grading/digital testing, AI career counselling · 3. Smart systems & connected devices design · 4. Primary | **5-mark:** Information Sources.

Impacts of Computing & Assistive Technologies

Impacts in 5 Fields · Assistive Technologies · Importance · Careers

★ **Must lock:** Assistive Technologies

5.4 Impacts of Computing in Various Fields

Computing ne zindagi tez aur connected ki — magar privacy risk aur technology addiction jaise masail bhi. **Har field ke dono rukh** socho.

Field	Positive (+)	Negative (–)
Education	Online classes, e-books, apps — flexible, accessible	Zyada screen time: attention kam, aankh/kamar dard
Communication	Email, video call — foran, global, takreeban muft (Dubai mein rishtedar ko WhatsApp call)	Aamne-saamne mulaqat kam → social isolation
Business	Online shopping, digital banking, automation — speed	Manual jobs machines ko; banks pe cyber-attacks
Healthcare	Digital records, telemedicine — door mareez ka ilaj	Hospital server hack → patient data privacy khatre mein
Agriculture	Smart irrigation, weather forecast — paidawar zyada	Dehaat mein internet kam → kisan use nahi kar pate

🧠 **Yaad rakho:** "**E-C-B-H-A**" — aur har ek ke liye ek + aur ek – zaroor.

5.5.1 Importance of Assistive Tech

Equal Access → Independence → Inclusion

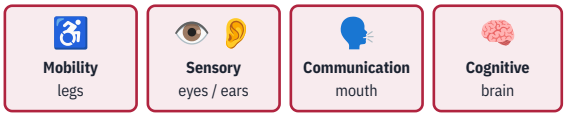
Wajah	Misaal
Equal access to education	Screen reader se nabina student bhi digital textbook parhe, classmates ki tarah
Independence in daily life	Smart wheelchair — har waqt kisi ki madad ki zaroorat kam
Social **inclusion**	Rukawatein khatam; aitmaad se kaam, parhai, mel-jol

🧠 **Yaad rakho:** "**AII**" = Access, Independence, Inclusion.

5.5 Assistive Technologies ★ Golden

Def (paper mein yehi likho): *Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering).*

Inse zyada log school, kaam aur muashre mein **barabar** hissa le sakte hain. Chaar categories:



Category	Maqsad	Misaalein
1. Mobility aids	Ek jagah se doosri jagah jana	Smart wheelchairs, stair lifts
2. Sensory aids	Dekhne/sunne mein madad	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Bolne mein mushkil walon ka izhar	Speech-generating devices, text-to-speech apps
4. Cognitive / learning	Yaad-dasht, parhna, samajhna	Audiobooks, spell-checkers, reminder apps

🧠 **Yaad rakho:** "**M**y **S**mall **C**at **C**ries" = Mobility, Sensory, Communication, Cognitive → taangein, aankh/kaan, munh, dimagh.

📝 **Paper mein:** "What are assistive technologies? Explain categories with examples" (long) · "Do sensory aids?" · "Screen reader is an example of ___" → sensory aid.

5.5.2 Careers in Assistive Tech

Career	Kaam
Assistive Software Developer	Speech-to-text tools (Urdu voice typing)
Biomedical Engineer	Artificial limbs, hearing aids design
UX/UI Accessibility Expert	Websites nabina users ke liye aasaan (images ke text descriptions)

5-mark frame: Assistive Technologies

Define	Devices, software and systems that help people with disabilities or functional limitations.
Explain	4 categories — mobility, sensory, communication, cognitive.
Example	Smart wheelchair, screen reader, text-to-speech, audiobooks.
Diagram	4 boxes (taangein, aankh/kaan, munh, dimagh).
Working	Importance: Access, Independence, Inclusion (AII).

Checkpoint: (1) Healthcare mein computing ka ek + aur ek –? (2) Hearing aid kis category mein? (3) Assistive tech ki ahmiyat ki 3 wajah? (4) Artificial limbs kaun design karta hai?

Jawab: 1. + Digital records/telemedicine; – hack se privacy risk · 2. Sensory aids · 3. Equal access, independence, inclusion · 4. Biomedical engineer | 60% se kam → kal Golden repair.

Day 20

Digital Literacy, Types of Data & Collection Strategies

Digital Literacy · Information Age · Qualitative vs Quantitative · 5 Strategies · Interviews & Surveys

★ **Must lock:** Types of Data, Strategies, Interviews & Surveys

6.1 Digital Literacy

Def (paper mein yehi likho): *Digital literacy is the ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner.*

Locate info → Organise → Create (useful cheez) → Share responsibly

Lafz	Matlab	Misaal
Safe	Apni aur doosron ki hifazat	Password / private details share nahi
Effective	Kaam achhe se ho	Sahih info jaldi dhoondna
Responsible	Imaandari aur insaaf	Copy nahi, fake news nahi phailana

🧠 **Yaad rakho:** Chaar verbs = "**LOCS**" (Locate, Organise, Create, Share) — ek **lock** 🔒 jo digital zindagi mehfooz rakhe.

6.2 Information Age — "Tech, Think, Talk"

Information Age = maujooda daur jahan internet, mobile aur platforms (websites, apps, social media) se bohat sa data foran banta, share aur access hota hai. Smartphone hona kaafi nahi — info samajhne, sambhalne aur samajhdari se use karne ki **skills** chahiye.

Ability	Matlab	Misaal
Technical	Devices & software chalana	Google Form banana, spreadsheet
Cognitive	Info pe tanqeedi soch	News sach hai? pehle check
Social	Online achha bartao	Class WhatsApp group mein adab, privacy

6.3 Types of Data ★ Golden

Def (paper mein yehi likho): *Qualitative data is descriptive and non-numerical; quantitative data is numerical, countable or measurable.*

	Qualitative	Quantitative
Nature	Descriptive, alfaaz	Numbers, gin/naap sakein
Examples	Opinions, reasons, behaviour	Hours, scores, %, totals
Best for	**Kyun / kaise** samajhna	Compare + graphs/charts

🧠 **Yaad rakho:** **QUAL**itative = **QUAL**ity = alfaaz/jazbaat. **QUANT**itative = **QUANT**ity = ginti. "Time online in hours" → quantitative.

6.4 Data-Collection Strategies ★ Golden

Def (paper mein yehi likho): *A data-collection strategy is a planned method for gathering original (new) information.*

Strategy	Kaise kaam kare	Data type	Kab best
Interview	Ek/chhote group se planned guftagu	Mostly qualitative	Kam logon se gehre jawab
Survey	Same sawal bohat logon se	Quant + qual	Bohat logon se jaldi
Prototype	Users early version try kar ke feedback	Dono	Product behtar karna
Observation	Dekho aur record karo, sawal nahi	Notes / counts	Log asal mein kya karte hain
Simulation	Computer model, variables badlo	Mostly quant	Asal tajurba mehenga/khatarnak

🧠 **Yaad rakho:** "**I** **S**aw **P**eople **O**n **S**creens" = Interviews, Surveys, Prototypes, Observation, Simulations. **Paper:** paanchon + definition + misaal + data type.

6.4.1–6.4.2 Interviews & Surveys ★ Golden

Interview = kisi shakhs se **planned guftagu** — **In-depth with few** (one-to-one ya chhota group). Sawal pehle se tayyar; jawab zyada tar qualitative.

Survey = **Same questions to many** (census jaisa) — Google Forms, Microsoft Forms.

Question type	Misaal	Data
Closed: Yes/No	Parhte waqt phone paas hota hai? (Yes/No)	Quantitative
Multiple choice	Phone zyada kis liye? (Lectures/Notes/Social/Games)	Ginne wali categories
Number	Roz kitne ghante phone? ___	Quantitative
Rating scale	"Phone parhai mein madad karta" 1 (strongly disagree) … 5	Quant (average)
Open-ended	Phone parhai mein kaise madad karta hai?	Qualitative

⚠ **Dhoka:** Biased sawal: "*Kya aap nahi mante phone waqt zaya karta hai?*" → neutral: "*Phone aap ke parhai ke waqt pe kya asar dalta hai?*" Do baatein ek sawal mein mat poochho.

Drill: (1) Digital literacy ki definition? (2) Information Age? (3) "Time online in hours" — qual ya quant? (4) Paanch strategies? (5) Survey ke do digital tools?

Jawab: 1. Safe, effective, responsible tareeqe se digital devices, internet, tools samajhna/use karna · 2. Daur jahan data foran banta/share hota hai · 3. Quantitative · 4. Interviews, surveys, prototypes, observation, simulations · 5. Google Forms, Microsoft Forms | **5-mark:** Data-collection strategies.

Day 21

Prototypes, Observation, Primary/Secondary & Spreadsheets

Prototypes · Observation · Simulations · Primary vs Secondary · Designing an Approach · Spreadsheets & Presentations

★ **Must lock:** Prototypes/Observation/Simulations, Primary & Secondary Data

6.4.3–6.4.5 Prototypes, Observation, Simulations ★ Golden

Method	Definition	Misaal
Prototype	Product/solution ka saada early version; final se pehle idea test. Users ka feedback = data	School app ki screens kaghaz pe bana ke 5 students se try karwao
Observation	Logon ko dekh ke record karna — koi sawal nahi; jo log **asal mein** karte hain	Library mein ginna kitne students phone use karte hain
Simulation	Computer model mein variables badal ke natija dekhna	Car crash test, traffic flow, flight simulator

🧠 **Yaad rakho:** **Dekho** → Observation · **Poochho** → Survey/Interview · **Try karwao** → Prototype · **Model banao** → Simulation.

📝 **Paper mein:** Scenario: "Which method is most suitable for ...? Give a reason." Method + uski khaas baat se wajah.

6.5 Primary vs Secondary Data ★ Golden

Aspect	Primary	Secondary
Source	Researcher ne **khud** collect kiya	Doosron ne pehle se collect kiya
Examples	Survey responses, interviews, observations	Reports, articles, websites, databases
Purpose	Khaas study ke liye tayyar	Naye maqsad ke liye dobara use
Control	**High** — sawal aur tareeqa apna	**Limited** — data pehle se maujood

🧠 **Yaad rakho:** Primary = "1st hand" (aap ne pehle collect kiya). Secondary = "2nd hand" — purani kitaab jaisa, pehle kisi aur ne use kiya.

📝 **Paper mein:** Sab se zyada poocha jane wala: 4-row table — Source, Examples, Purpose, Control.

6.7.2 Presentations

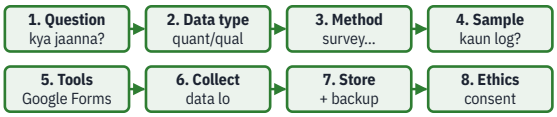
Live audience ko findings batane ke liye. Rule: **ek slide = ek idea**, ~6 chhote bullets max, chart slide pe insert.

Spreadsheet data → Chart → Insert → Slides

⚠ **Dhoka:** Slide se lambe jumle parh rahe ho? Slide mein zyada text hai.

6.6 Designing a Data-Collection Approach

Data lene se **pehle plan**:



1. Research question saaf likho — kya jaanna hai?
2. Data type: quantitative, qualitative ya dono
3. Method chuno (survey, interview, observation)
4. Sample / target group — kaun data dega
5. Tools (Google Forms, spreadsheet)
6. Data collect karo
7. Mehfooz store + **backup**
8. Ethics: consent, privacy

🧠 **Yaad rakho:** "**Q**uiet **T**eachers **M**ake **S**tudents **T**hink, **C**heck, **S**ave **E**thically".

6.7.1 Spreadsheets — Formulas

Chaar tools: spreadsheets, presentations, infographics, reports. Spreadsheet = **hisaab**.

Formula	Kaam
`=SUM(B2:B4)`	B2 se B4 tak jama
`=AVERAGE(B2:B4)`	Mean $\bar{x} = \frac{\sum x}{n}$
`=MAX(B2:B4)` / `=MIN()`	Sab se bari / chhoti
`=MEDIAN(B2:B4)`	Sort kar ke beech wali value
`=COUNT(B2:B4)`	Number wale cells gino
`=COUNTA(C2:C4)`	Khali na hon (text ya number)
`=COUNTIF(C2:C4,"Yes")`	Shart poori karne wale cells

Worked: hours = 3, 5, 2 → SUM 10, AVERAGE $\frac{10}{3} \approx 3.33$, MEDIAN 3 (2, **3**, 5), MAX 5.

⚠ **Dhoka:** Formula hamesha **=** se shuru. Range colon se: `B2:B4` (comma nahi). Text shart quotes mein: `"Yes"`.

5-mark frame: Data-Collection Strategies

Define	A data-collection strategy is a planned method for gathering original information.
Explain	5 strategies: interview, survey, prototype, observation, simulation.
Example	Survey: 40 students se phone hours; observation: library mein ginti.
Diagram	5 boxes: few-deep / many-quick / try / watch / model.
Working	Har ek ka data type + kab best (table).

Drill: (1) Engineer simulation kyun use kare? (2) Kis data mein researcher ka control high? (3) Design ka pehla step? (4) D2 se D20 ka average formula?

Jawab: 1. Asal tajurba mehenga, khatarnak ya mushkil · 2. Primary · 3. Research question saaf define karo · 4. `=AVERAGE(D2:D20)` | **5-mark:** Primary vs Secondary data.

Day 22

Infographics, Reports & Digital Inquiry (Steps 1–6)

Infographics & Reports · Case Study · Advanced Search · Survey Preparation & Ethics

★ **Must lock:** Case Study, Steps 1–3, Steps 4–6

6.7.3–6.7.4 Infographics & Reports

Infographic (information + graphic) = info/data ka visual roop taake asal baat foran samajh aaye: mushkil concept saada karo, ahem findings highlight, kam text, sahih info. Tools: **Canva, Power BI, Tableau**. **Report** = tafseeli, structured document (method + recommendations).

Situation	Best tool	Kyun
40 students ke average marks	Spreadsheet	Formulas foran hisaab
Assembly mein class ko results batana	Presentation	Live audience ke liye
Notice board ke liye ek page poster	Infographic	Ek nazar mein samajh
Principal ko poori study	Report	Tafseeli, structured

🧠 **Yaad rakho:** Spreadsheet = **Hisaab**, Presentation = **Batao**, Infographic = **Jaldi dikhao**, Report = **Poora samjhao**.

6.8 Steps 1–3: Advanced Search ★ Golden

Pehle search engine (Google, Bing) se background info, **advanced search operators** se results tang aur reliable:

Query	Operator	Kaam
`"mobile phone use" "study habits"`	`" "` quotes	Exact phrase, bikhre alfaaz nahi
`"academic performance" filetype:pdf`	filetype:	Sirf PDF (reports, research papers)
`digital distraction site:.edu`	site:	Sirf educational websites

Step 2: data types tay karo (hours = quant, reasons = qual). **Step 3**: methods chuno — survey + secondary articles.

🧠 **Yaad rakho:** Teen operators, teen kaam: **" "** = exact, **filetype:** = format, **site:** = kahan. Colon ke baad **space nahi** (`site:.edu`).

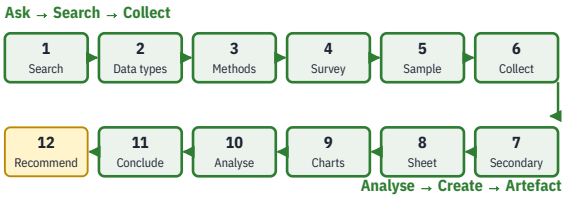
📝 **Paper mein:** "Explain any three advanced search operators with examples" — operator, misaal query, kya karta hai.

5-mark frame: Digital Inquiry Project

Define	An inquiry is an investigation to find the answer to a question using digital tools.
Explain	12 steps: search → collect → analyse → create.
Example	Mobile phone use vs study habits (40 students).
Diagram	12-step snake flow / 4 stages.
Working	Survey + article → spreadsheet → chart → conclusion → artefact.

6.8 Case Study: Digital Inquiry Project ★ Golden

Inquiry = kisi sawal ka jawab dhoondne ki tehqeeq. **Research question:** "*How does daily mobile phone use affect the study habits of students?*" — kuch kehte phone se seekhte, kuch kehte waqt zaya.



Stage	Steps
Question & Search	1 Advanced searches · 2 Data types · 3 Methods
Collect	4 Survey tayyar · 5 Sample · 6 Primary data ethically · 7 Secondary data
Analyse	8 Spreadsheet · 9 Charts · 10 Analyse
Create	11 Conclusion · 12 Recommendations → final digital artefact

🧠 **Yaad rakho:** Paanch alfaaz: **Ask → Search → Collect → Analyse → Create**. Har research isi pattern pe.

📝 **Paper mein:** Long Q: "Describe the steps of a digital inquiry project (mobile phone example)" — pehle research question, phir 12 steps.

6.8 Steps 4–6: Survey & Primary Data ★ Golden

Survey question	Data
Roz kitne ghante phone use karte ho?	Quantitative
Phone sab se zyada kis kaam ke liye?	Quant (categories)
Phone parhai mein kaise madad karta hai?	Qualitative
Phone zimmedari se use karne ka ek tareeqa batao	Qualitative

Sample = woh group jo data deta hai (yahan 40 students). **Ethics:**

- **C**onsent — ijazat lo, maqsad batao
- **N**o sensitive — zaati/sensitive sawal nahi
- **A**nonymous — naam nahi
- **P**rivate — data mehfooz, share nahi
- **E**ducational — sirf taleemi maqsad

🧠 **Yaad rakho:** "**C-NAPE**" — banday ki gardan (nape) dhaanp ke rakho = us ki hifazat.

Drill: (1) Infographic ke do tools? (2) Case study ka research question? (3) "online learning" exact phrase sirf educational sites pe — query? (4) Sample kya hai?

Jawab: 1. Canva, Power BI (Tableau) · 2. How does daily mobile phone use affect the study habits of students? · 3. `"online learning" site:.edu` · 4. Data dene wala logon ka group | **5-mark:** Digital Inquiry Project.

Day 23

Inquiry Steps 7–12, Final Artefact & Careers

Secondary Data · Spreadsheet · Charts · Analysis · Conclusion · Recommendations · Digital Artefact · Careers

★ **Must lock:** Steps 7–9, Steps 10–12, Final Artefact

6.8 Steps 7–9: Secondary Data & Spreadsheet ★ Golden

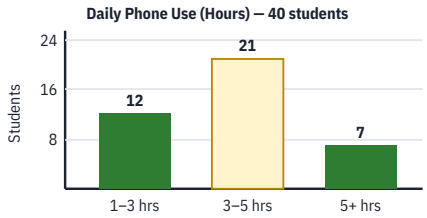
Step 7: reliable online article — "*excessive screen time lowers concentration*" (secondary) ko apne survey (primary) se milao. Mutfiq → conclusion mazboot. Ikhtilaf → wajah socho (umar ka group, mulk).

Step 8: spreadsheet mein organise:

A: Student	B: Hours	C: Study Use	D: Activity	E: Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

Total `=SUM(B2:B4)` → 10 · Average `=AVERAGE(B2:B4)` → 3.33 · Study users `=COUNTIF(C2:C4,"Yes")` → 2

Step 9: chart banao:



Counts (12, 21, 7) source chart ki bars se andaza · total 40

🧠 **Yaad rakho:** Chart chooser: **Bar** = compare, **Pie** = whole ke hisse, **Line** = waqt ke saath badlao.

📝 **Paper mein:** Table de ke: "total/average hours ka formula likho" ya "study ke liye use karne wale gino" — poora formula range ke saath.

Careers in Digital Research

Career	Kaam	Skill
Market Research Analyst	Business decisions ke liye surveys	Survey, primary data
UX Researcher	App/website design behtar karne ko data	Prototype, interview, observation
Data Journalist	Search se stories, infographics se pesh	Advanced search, infographic
Digital Marketing Specialist	Audience data → dashboards, reports	Dashboard, report

6.8 Steps 10–12: Analysis, Conclusion, Recommendations ★ Golden

Analyse = data ko ghaur se parh ke patterns aur trends dhoondna:

- Zyada tar students (**21/40** $\approx 52\%$) roz **3–5 ghante** phone use karte hain.
- **Aadhe** taleemi kaam (lectures/notes) ke liye.
- Trend: jo zyada **entertainment** ke liye use karte, un mein distraction bohat zyada.

$$\text{Percentage} = \frac{\text{part}}{\text{total}} \times 100 = \frac{21}{40} \times 100 = 52.5\%$$

Conclusion formula: "The data **shows** … **However** … **Therefore** …"

Def (paper mein yehi likho): The data shows most students use phones 3–5 hours daily. However, half use them for lectures and notes. Therefore, phones help study when used for learning, but entertainment use increases distraction.

Recommendations (har ek finding se nikle): parhai ke waqt notifications band / focus mode; entertainment ka waqt mehdood; teachers achhi study apps batayein.

📝 **Paper mein:** "Write a conclusion and two recommendations" — shows/however/therefore + har recommendation kisi finding se jurri ho.

6.8 The Final Digital Artefact ★ Golden

Def (paper mein yehi likho): *A digital artefact is a final product created using digital technologies that answers the research question with evidence and conclusions.*

Sirf sajaya hua file nahi — **saboot aur natija** dikhaye.

Option	Tools	Maqsad
Digital Report	MS Word, Google Docs	Poori study structured text mein
Presentation	PowerPoint, Canva, Slides	Live audience ko key findings
Infographic	Canva, Adobe Express	Ek page pe results jaldi
Dashboard	Power BI, Tableau	Interactive charts & filters

What: title, question, intro → **How**: method → **Found**: primary, secondary, charts → **So what**: conclusion, recommendations, references

🧠 **Yaad rakho:** Career ↔ tool: Market Researcher → Survey; UX → Prototype; Journalist → Search + Infographic; Marketer → Dashboard.

Checkpoint: (1) Step 7 ka secondary data? (2) Step 10: kis group mein distraction zyada? (3) Interactive filters wala artefact? (4) Business ke liye surveys kaun design karta hai?

Jawab: 1. Reliable article: "excessive screen time lowers concentration" · 2. Jo entertainment ke liye use karte · 3. Dashboard (Power BI, Tableau) · 4. Market Research Analyst | 60% se kam → kal Golden repair.

Day 24

Golden Blitz — Chapters 1 & 2

Sirf ★ cards · 14 Golden · har table aur diagram yaad se likho, phir check

★ **Must lock:** 14 Golden cards (Ch 1–2)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Analog vs Digital — Ch1 · 1.1.3

A signal is a changing voltage that carries information.



Analog: continuous, sine, noise se asar, voice. Digital: 2 levels (1/0), square, noise-resistant, computer data.

☐ Logic Gates AND/OR/NOT — Ch1 · 1.1.7



AND

$$Y = A \cdot B$$

: 1 sirf 11 · OR

$$Y = A + B$$

: 0 sirf 00 · NOT

$$Y = \bar{A}$$

. NAND/NOR = universal (+ bubble).

AB	AND	OR	NAND	NOR	XOR
00	0	0	1	1	0
01	0	1	1	0	1
10	0	1	1	0	1
11	1	1	0	0	0

☐ Logic Diagrams — Ch1 · 1.1.11

Picture of a Boolean expression: how gates connect.

Precedence: ( ) → NOT → AND → OR. Aakhri operation ka gate Y ke paas.

$$Y = A \cdot B + C$$

= AND → OR (2 gates).

☐ K-Maps — Ch1 · 1.1.13

Graphical tool to simplify Boolean expressions; adjacent cells differ by one bit (Gray code).

Columns 00, 01, 11, 10 · groups 1/2/4/8 · wrap-around · badalne wala variable gira do.

$$\sum m(0,1,2,3,7) = \bar{A} + BC$$

☐ SDLC — Ch1 · 1.2.1

Structured process that guides software development from beginning to end.

Requirement (SRS) → Design → Implementation → Testing → Deployment → Maintenance · "R-D-I-T-D-M"

☐ Waterfall Model — Ch1 · 1.2.4

Linear, sequential; each phase completes before the next.

+ simple, documented · – inflexible, late testing · fixed requirements pe use.

☐ Agile Model — Ch1 · 1.2.4

Iterative; small cycles called sprints (2–4 weeks).

Plan → Design → Build → Test → Review ↺ · continuous feedback · + flexible, early delivery · – kam docs, cost unpredictable.

"Requirements badlengi?" Haan → Agile, Nahi → Waterfall.

☐ OSI 7 Layers — Ch1 · 1.3

ISO, 1984. 1 Physical · 2 Data Link · 3 Network · 4 Transport (TCP/UDP) · 5 Session · 6 Presentation (encryption) · 7 Application (HTTP)

Please Do Not Throw Sausage Pizza Away · TCP/IP 4 = 3-1-1-2

TCP/IP	OSI layers
Application	7, 6, 5
Transport	4
Internet	3
Network Access	2, 1

☐ Algorithm — Ch2 · 2.2

A step-by-step logical procedure used to solve a problem.

Input · clear steps · definite end · output. Importance **S-O-R-A-B**: Structured, Optimised, Reusable, Accurate, Blueprint.

☐ Algorithm vs Pseudocode — Ch2 · 2.2.2

Algorithm = **what**, language-independent, informal, sab ke liye. Pseudocode = **how**, IF/FOR/WHILE, standard conventions, programmers ke liye, executable nahi.

☐ Decomposition — Ch2 · 2.3.1

Breaking a complex task into smaller, manageable sub-tasks.

Complex problem → sub-task 1, 2, 3 → solve alag → jodo. "Roti todo"

English seekhna: vocabulary → grammar → tenses → sentences.

☐ Bubble Sort — Ch2 · 2.4.1

Repeatedly compares adjacent elements and swaps if in wrong order.

[8,4,1,9,3] pass 1 → 4,1,8,3,**9**. Passes

$$N - 1$$

, comparisons

$$\frac{N(N-1)}{2}$$

. Stable.

☐ Selection Sort — Ch2 · 2.4.1

Repeatedly selects the smallest element and moves it to the front.

Fixed DATA[K] vs outer+1 se aage. Pass 1 → **1**,8,4,9,3. Stable nahi.

Pass	Bubble	Selection
1	4,1,8,3,9	1,8,4,9,3
2	1,4,3,8,9	1,3,8,9,4
3	1,3,4,8,9	1,3,4,9,8
4	1,3,4,8,9	1,3,4,8,9

☐ Binary Search — Ch2 · 2.4.2

Repeatedly divides the search interval in half — list must be sorted.

$$MID = \lfloor \frac{BEG+END}{2} \rfloor$$

; chhota → END=MID−1; bara → BEG=MID+1; while BEG ≤ END. 1..9 mein 7: MID 5 → 7 ✔ (2 steps).

$$O(\log N)$$

#	BEG	END	MID	Action
1	1	9	5	7>5 → BEG=6
2	6	9	7	Found

Blitz score

Ch1 ☐☐☐☐☐☐☐☐ / 8

Ch2 ☐☐☐☐☐☐ / 6

Golden Blitz — Chapters 3 & 4

Sirf ★ cards · 16 Golden · code aur tables yaad se likho

★ Must lock: 16 Golden cards (Ch 3–4)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Python Fundamentals — Ch3 · 3.4

Statements top-to-bottom. # comment. **Indentation** (4 spaces) = block (C++/Java { }). : ke baad block.

☐ Variables — Ch3 · 3.4.1

A named space in memory used to store data.

Letters/digits/_ · digit se shuru nahi · case-sensitive · keyword nahi. ✔ _total, marks2026 ✘ 2marks, roll-no, class

☐ Data Types — Ch3 · 3.4.1

Type	Example	Note
int	16	whole
float	5.7	decimal
str	"2026"	quotes
bool	True	capital T/F
list	[85, 90]	changeable
tuple	(10, 20)	locked
dict	{"age": 16}	key:value
set	{'a','e'}	unique

☐ Arithmetic & Assignment — Ch3 · 3.4.3

9/2=4.5 · 9//2=4 · 9%2=1 · 2**3=8 · x+=3 = x=x+3. Precedence: () → ** → * / // % → + −

☐ Relational & Logical — Ch3 · 3.4.3

== != > < >= <= → True/False. and (dono), or (koi ek), not (ulta). Order not → and → or. `in`, `not in`.

☐ Bitwise — Ch3 · 3.4.3

12 & 10 = 8 · 12 | 10 = 14 · 12 ^ 10 = 6 ·

$$\sim x = -(x+1)$$

.

$$x \ll n = x \cdot 2^n$$

.

$$x \gg n = \lfloor x/2^n \rfloor$$

☐ Control Structures — Ch3 · 3.5

Decide the order in which statements are executed.

S-S-R: Sequence (default), Selection (if), Repetition (for/while).

☐ if & if-else — Ch3 · 3.5.2

```
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Colon + indent; else ki condition nahi.

☐ if-elif-else — Ch3 · 3.5.2

Upar se neeche; pehli True chale, baqi skip. Sakht condition pehle (≥80 → ≥70 → ≥60 → else). Nested if = if ke andar if.

☐ for & range() — Ch3 · 3.5.3

Ginti maloom → for. range(3)=0 1 2 · range(2,5)=2 3 4 · range(1,6,2)=1 3 5 · ek pehle ruke.

```
total = 0
for i in range(1, 11):
    total += i
print(total)   # 55
```

☐ while, break, continue — Ch3 · 3.5.3

Condition True tak. **I-T-U**: Initialise, Test, Update (warna infinite). break = loop khatam; continue = agli iteration.

```
count = 1
while count <= 3:
    print(count)
    count += 1   # 1 2 3
```

range(1,6): break at 3 → 1 2 · continue at 3 → 1 2 4 5

☐ Nested Loops, for vs while — Ch3 · 3.5.3

Inner har outer iteration pe poora: runs = outer × inner (2×3=6). for = known count, auto update; while = unknown, manual update.

☐ Keys & Integrity — Ch4 · 4.3.1

A key uniquely identifies records and links tables.

PK	FK
unique, not null, sirf ek	doosri table ki PK refer; duplicate/null ho sakti

Candidate, alternate, composite.

☐ ER Model — Ch4 · 4.4

▭ Entity · ⬭ Attribute (PK underline) · ◇ Relationship · 1:1, 1:M, M:N → M:N ko junction table se do 1:M.

Library: MEMBER 1—M **BorrowRecord** M—1 BOOK (MemberID FK, BookID FK).

☐ Referential Integrity — Ch4 · 4.5

Every FK value in the child must match an existing PK in the parent (or be NULL).

Orphan nahi. Cascade Update / Cascade Delete (khatarnak).

☐ Queries in MS Access — Ch4 · 4.11

Create → Query Design → Show Table → fields → Criteria → Run (!). AND = **same row**, OR = **or: row**. Text "IT" quotes mein; >=80. Totals Σ: Group By, Count, Sum, Avg.

Blitz score

Ch3 ☐☐☐☐☐☐☐☐☐☐☐☐ / 12

Ch4 ☐☐☐☐ / 4

Day 26

Golden Blitz — Chapters 5 & 6

Sirf ★ cards · 15 Golden · definitions word-for-word bolo

★ **Must lock:** 15 Golden cards (Ch 5–6)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Artificial Intelligence Ch5 · 5.1.1

Ability of a computer system to perform tasks that normally need human thinking.

Face unlock, YouTube recommendations, e-challan. IoT (eyes) → Analytics → AI (brain) → Actuator.

☐ IoT & Components Ch5 · 5.1.2

Physical devices connected to the Internet that collect and share data to perform tasks automatically.

Sensor → Connectivity → Data processing → Cloud → Actuator → UI.

Component	Example
Sensor	temperature, soil moisture
Actuator	motor, water valve
Connectivity	Wi-Fi, Bluetooth, 4G
Processing	garam? → AC on
Cloud	Google Drive, iCloud
UI	mobile app, voice

Sensor senses, actuator acts.

☐ Information Sources Ch5 · 5.3

Primary = firsthand (interview, NADRA records, photo) · Secondary = explained (newspaper, textbook) · Tertiary = summary (encyclopedia, Wikipedia). Author–Date–References check.

☐ Assistive Technologies Ch5 · 5.5

Devices, software and systems that help people with disabilities.

Mobility (wheelchair) · Sensory (screen reader, hearing aid) · Communication (TTS) · Cognitive (audiobooks). Importance: AII.

☐ Types of Data Ch6 · 6.3

Qualitative = alfaaz, kyun/kaise · Quantitative = numbers, compare + charts. "Hours online" = quant.

☐ Data-Collection Strategies Ch6 · 6.4

A planned method for gathering original information.

Interview · Survey · Prototype · Observation · Simulation — "I Saw People On Screens".

Strategy	Best when
Interview	gehre jawab, kam log
Survey	bohat log, jaldi
Prototype	product behtar
Observation	asal behaviour
Simulation	mehenga/khatarnak

☐ Interviews & Surveys Ch6 · 6.4.1

Interview = in-depth with few (qual). Survey = same questions to many (Google/MS Forms). Question types: Yes/No, MCQ, number, rating 1–5, open-ended.

☐ Prototype · Observation · Simulation Ch6 · 6.4.3

Try it (early version + feedback) · Watch it (no questions) · Model it (risky/mehenga experiment ki jagah).

☐ Primary vs Secondary Data Ch6 · 6.5

	Primary	Secondary
Source	khud	doosre
Control	high	limited
Ex.	survey	reports

☐ Digital Inquiry Project Ch6 · 6.8

Question: phone use vs study habits. Ask → Search → Collect → Analyse → Create · 12 steps → artefact.

☐ Steps 1–3: Advanced Search Ch6 · 6.8

`" "` exact · `filetype:pdf` format · `site:.edu` kahan (colon ke baad space nahi).

☐ Steps 4–6: Survey & Ethics Ch6 · 6.8

Clear, unbiased sawal; sample = 40 students. Ethics **C-NAPE**: Consent, Non-sensitive, Anonymous, Private, Educational.

☐ Steps 7–9: Secondary & Sheet Ch6 · 6.8

Article "screen time lowers concentration" vs survey. `=SUM(B2:B4)`, `=AVERAGE()`, `=COUNTIF(C2:C4,"Yes")`. Bar chart 12 / 21 / 7.

☐ Steps 10–12: Conclusion Ch6 · 6.8

Majority 3–5 hrs; aadhe study; entertainment → distraction. "Shows … However … Therefore …" + recommendations finding se.

☐ Final Digital Artefact Ch6 · 6.8

Report (Word/Docs) · Presentation (PowerPoint/Canva) · Infographic (Canva) · Dashboard (Power BI/Tableau). What → How → Found → So what.

Blitz score

Ch5 ☐☐☐☐ / 4

Ch6 ☐☐☐☐☐☐☐☐☐☐☐ / 11

Day
27

Night-before Recaps — Saare Chapters

Kitaab band karo · har recap ounchi awaaz mein bolo · jo na bol sako, mark karo

★ **Must lock:** All chapters

Ch 1 Computer Systems

- Discrete = gina (stairs); continuous = naapa (ramp). Digital: 0/1 — reliable, accurate, easy, programmable.
- Analog = sine, smooth; digital = square HIGH/LOW, noise-resistant.
- AND · 1 sirf sab 1 · OR + 1 agar koi 1 · NOT ' · NAND/NOR universal · XOR 1 jab alag · XNOR 1 jab same.
- Rows $= 2^n$. SOP = minterms (1s), POS = maxterms (0s). K-map groups 1/2/4/8, wrap-around.
- SDLC 6 phases; Waterfall sequential; Agile sprints + feedback.
- OSI 7 (Please Do Not Throw Sausage Pizza Away); TCP/IP 4.

Ch 2 Computational Thinking & Algorithm

- CT: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps; pseudocode = structured English.
- Bubble: parosi swap, bari value aakhir. Selection: minimum aage.
- Linear: ek ek, unsorted bhi. Binary: sorted ZAROORI, MID test, low/mid/high trace.
- Judge: correctness, efficiency, clarity. Bara sorted → binary.

Ch 3 Programming Fundamentals

- Python high-level, interpreted. int, float, str, bool, list, dict. input() → str.
- + - * / // % ** · relational · and or not · & | ^ ~ << >>.
- if / if-else / elif / nested — pehli True jeetay.
- for = ginti maloom + range; while = update zaroor; break nikle, continue skip.
- math, random, datetime; pip: NumPy, Pandas, Matplotlib.
- Syntax = chale hi nahi · Runtime = crash · Logical = ghalat jawab.

Ch 4 Data and Analysis

- Data raw; information processed. DBMS: redundancy kam, security, sharing, integrity.
- Table/relation · row/record/tuple · column/field/attribute. PK unique; FK doosri PK.
- ER: entity ▭, attribute ⬭, relationship ◇; 1:1, 1:M, M:N.
- Referential integrity: orphan nahi; Cascade Update/Delete.
- Access: tables, forms, queries, reports. Criteria, wildcards * ?, AND/OR, Totals.

Ch 5 Application & Impacts

- IoT 6 parts: sensor, actuator, connectivity, processing, cloud, UI.
- IoT (eyes) → Analytics (sort) → AI (brain) → actuator.
- Sources: primary, secondary, tertiary; author/date/references nahi → trust nahi.
- Assistive: mobility, sensory, communication, cognitive → access, independence, inclusion.

Ch 6 Digital Literacy

- Find, evaluate, create, share — safely.
- Qual = alfaaz, quant = numbers. 5 strategies: interview, survey, prototype, observation, simulation.
- Primary khud, secondary pehle se. Survey sawal clear, ek idea, unbiased.
- Spreadsheet (=AVERAGE), slides, infographic, report. Sources cite karo. 12-step inquiry → artefact.

Mnemonic wall 🧠

R-D-I-T-D-M · Please Do Not Throw Sausage Pizza Away · 3-1-1-2 · Don't Pick Any Apple · S-O-R-A-B · C-E-C · S-D-A-W-A-C · E-S-A-R-D · S-S-R · I-T-U · Cows Sit Calmly And Vote · CHIEF H · PNEP · E-C-B-H-A · My Small Cat Cries · AII · LOCS · I Saw People On Screens · C-NAPE · Shows · However · Therefore

Paper ke pakke sawal 🎯

- Analog vs digital · gates + truth tables · K-map · SDLC phases · Waterfall vs Agile · OSI 7 layers
- Algorithm vs pseudocode · Bubble/Selection passes · Binary search trace
- Valid variable names · data types · evaluate // % ** & | ^ ~ · grade / even-odd / loop programs · break vs continue
- PK vs FK · ER diagram · referential integrity · Access query grid
- IoT components · AI + IoT + analytics · information sources · assistive tech
- Qual vs quant · 5 strategies · primary vs secondary · search operators · ethics · conclusion

Aage ka plan: Day 28 — mock: 2 h 30 min · 75 marks · hall conditions. **Day 29** — key se mark karo, har missed ★ ki repair list. **Day 30** — sirf repair list dobara; raat 10 baje so jao; koi naya topic nahi.

Day 28

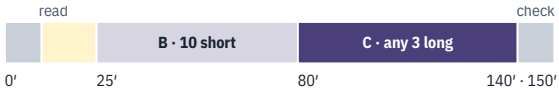
Mock Paper — Hall Conditions (1/2)

2 hours 30 minutes · 75 marks · no notes · Section A

★ **Must lock:** Whole syllabus

Aaj ke 3 ghante — Hall conditions

- **2 h 30 min** mein paper · **75 marks** · notes nahi, phone doosre kamre mein, Day 29 ke pages **mat dekho** (answers wahan hain — un pages ko alag rakh do).
- Bacha hua 30 min: paper pack karo — **aaj mark nahi karna**.
- Khud apne invigilator bano: timer lagao, ghari saamne, ek hi baith mein.



Time plan (minutes): pehle 10′ paper parho aur Section C ke 3 sawal chuno · A 15′ · B 55′ (~5′ har sawal) · C 60′ (20′ har sawal) · aakhri 10′ check

Paper strategy

- MCQ: pehle yaqeeni wale, shak wale pe nishaan, baad mein.
- Short (3 marks): **Define + 2 points/example**.
- Long (10 marks): **Define · Explain · Example · Diagram · Working**.
- Har diagram/table pe **label + title**.
- Python: sahih **indentation**, colon, quotes.

Mock paper — Computer Science XI

Time: 2 h 30 min · Total: 75 · Calculators allowed

Instructions: Answer all of Section A and B; attempt any **three** from Section C. Write Python with correct indentation. Draw neat, labelled diagrams.

Section A — Multiple choice

15 × 1 = 15

1. Which quantity is discrete?
   a. Room temperature b. Number of desks
   c. Car speed d. Time of day

2. NAND is called universal because:
   a. it has two inputs b. any gate can be built from it
   c. its output is always 1 d. it is faster than AND

3. The Boolean expression for XOR is:
   a. $A \cdot B$ b. $A + B$
   c. $A\bar{B} + \bar{A}B$ d. $(A \cdot B)'$

4. OSI layer that chooses the best path (IP addresses) is:
   a. Transport b. Network
   c. Data Link d. Session

5. Which SDLC model delivers working software in short sprints?
   a. Waterfall b. Agile
   c. Spiral only d. Big Bang

6. Binary Search requires the list to be:
   a. unsorted b. sorted
   c. all unique and unsorted d. of even length

7. Worst-case time of Bubble Sort is:
   a. $O(1)$ b. $O(\log n)$
   c. $O(n^2)$ d. $O(n)$

8. `input()` in Python returns a:
   a. int b. str
   c. float d. bool

9. `17 // 5` equals:
   a. 3.4 b. 3
   c. 2 d. 3.0

10. A program that runs but prints the wrong total has a:
    a. syntax error b. runtime error
    c. logical error d. hardware fault

11. A field that uniquely identifies each record is a:
    a. foreign key b. primary key
    c. composite description d. query

12. An actuator in IoT:
    a. collects temperature b. opens a water valve
    c. stores cloud files d. draws a chart

13. Wikipedia is a:
    a. primary source b. secondary source
    c. tertiary source d. sensor

14. Marks out of 100 are:
    a. qualitative data b. quantitative data
    c. tertiary data d. metadata only

15. Which Access object is best for printing a formatted class list?
    a. Table b. Form
    c. Query d. Report

MCQ answer grid — sahih harf pe gola lagao

1. ⓐ ⓑ ⓒ ⓓ	**2.** ⓐ ⓑ ⓒ ⓓ	**3.** ⓐ ⓑ ⓒ ⓓ	**4.** ⓐ ⓑ ⓒ ⓓ	**5.** ⓐ ⓑ ⓒ ⓓ	**6.** ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ
8. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
15. ⓐ ⓑ ⓒ ⓓ						

Kal (Day 29) isi grid ko key se milao — overwriting / do gole = ghalat.

Day 28

Mock Paper — Hall Conditions (2/2)

Section B (short) · Section C (long, any three)

★ **Must lock:** Whole syllabus

Section B — Short questions

10 × 3 = 30

1. Differentiate analog and digital signals with one example of each. [3]
2. Draw the symbol and truth table of a 2-input NOR gate. [3]
3. List the four pillars of computational thinking in one sentence each. [3]
4. Why can you not use Binary Search on an unsorted attendance list? [3]
5. Write a Python if-elif-else that prints Pass if marks ≥ 40, else Fail. [3]
6. Define primary key and foreign key. [3]
7. Name the six components of IoT. [3]
8. Classify: (a) a photograph you took, (b) a textbook, (c) an encyclopedia. [3]
9. Give two good and two bad survey questions about mobile-phone use in college. [3]
10. State three advantages of a DBMS over paper files. [3]

Section C — Long questions

Attempt any THREE · 3 × 10 = 30

1. Simplify
$$F(A, B, C) = \sum m(1, 3, 5, 7)$$
using a 3-variable K-map. Show the grouping and the final expression. Draw the logic diagram. [10]
2. Dry-run Bubble Sort on the list [5, 1, 4, 2]. Show every pass. Then search for 4 using Binary Search on the sorted result, with a low/mid/high table. [10]
3. Write a complete Python program that reads n subject marks, prints the average, and prints "A" if average ≥ 80, "B" if ≥ 70, "C" if ≥ 60, else "D". Include a sample run of four marks: 80, 70, 90, 60. [10]
4. Design an ER diagram and relational schema for a school library (Student, Book, Issue). Mark keys. Write one Access/SQL query that lists all books issued to roll number 101. [10]
5. Explain how IoT, Data Analytics and AI work together on a Sindh farm. Then discuss two positive and two negative impacts of computing on education in Pakistan. [10]

Rough-work / time log

Section	Shuru	Khatam	Kitne sawal
A			/15
B			/10
C			/3

Marks ↔ jawab ki lambai

Marks	Kitna likho
1 (MCQ)	Sirf sahih harf
3 (short)	Definition + 2 points / 4–6 lines ya ek chhota diagram
10 (long)	1½–2 pages: Define · Explain · Example · Diagram · Working

Section C — mera intikhab

Choice	Q no.	Kyun (strongest topic)
1st		
2nd		
3rd		

Paper khatam? 🛑

Kalam rakho. Paper ek lifafe mein band. **Aaj marking nahi** — kal fresh dimagh se Day 29 key ke saath. Raat ko bas aaram.

🧠 **Yaad rakho:** Long answer ka frame: **Define · Explain · Example · Diagram · Working** — kisi ek ke baghair poore marks nahi.

Day 29

Mark the Mock — Section A & B Key

0–90 min: key se mark karo · ghalat sawal ka Golden note karo

★ **Must lock:** Missed Goldens

⚠ **Dhoka:** Pehle **Day 28 ka paper poora** ho chuka ho — warna ye page mat parho. Jis jawab mein working, diagram ya example nahi, usay **poore marks mat do**.

Section A — key (15)

Q	Ans	Ghalat ho to repair	✔/✘
1	**b**	1.1.1 Discrete/continuous	☐
2	**b**	Universal gates (NAND/NOR)	☐
3	**c**	XOR / Logic gates	☐
4	**b**	★ OSI 7 layers	☐
5	**b**	★ Agile model	☐
6	**b**	★ Binary Search	☐
7	**c**	★ Bubble Sort / Big O	☐
8	**b**	Input & output (3.4.2)	☐
9	**b**	★ Arithmetic operators	☐
10	**c**	Debugging — error types	☐
11	**b**	★ Keys & integrity	☐
12	**b**	★ IoT & components	☐
13	**c**	★ Information sources	☐
14	**b**	★ Types of data	☐
15	**d**	Database objects (Report)	☐

A score: ____ / 15

Section B — model answers (10 × 3)

Q	Model answer (in points ko dhoondo)	Repair topic	Mine
1	Analog: continuous, sine wave — human voice. Digital: two levels 0/1, square wave — computer data. Digital noise-resistant.	★ Analog & digital	/3
2	 $$Y = \overline{A + B}$$. 00 → 1, 01 → 0, 10 → 0, 11 → 0.	★ Logic gates	/3
3	Decomposition = problem todo · Pattern recognition = milti-julti cheez reuse · Abstraction = faltu detail chhoro · Algorithm = tarteeb wale steps.	2.1 CT pillars	/3
4	Binary sorted range ko aadha karta hai; unsorted list mein MID batata hi nahi kaunsa half rakhein → method invalid.	★ Binary Search	/3
5	`m = int(input("Marks: ")) · if m >= 40: print("Pass") · else: print("Fail")`	★ if-else	/3
6	PK: row ki unique pehchan (StudentID). FK: doosri table ki PK ko refer (Issue.StudentID).	★ Keys	/3
7	Sensors, actuators, connectivity, data processing, cloud computing, user interface.	★ IoT components	/3
8	(a) primary (b) secondary (c) tertiary.	★ Information sources	/3
9	Good: "Roz study ke liye kitne ghante phone?" / "Class notes ke liye kaunsi apps?" Bad: "Kya aap nahi mante phone tabahi hai?" (leading) / "Phone pasand aur laptop bhi use?" (double-barrelled).	★ Interviews & surveys	/3
10	Kam redundancy · aasaan search/update · security + multi-user · integrity constraints (koi 3).	4.1.2 DBMS	/3

B score: ____ / 30 · Marking: 1 mark definition/sahih term, 1 mark explanation, 1 mark example/diagram.

Section A — har jawab ki wajah

Q	Kyun yahi jawab? (MCQ wazahat)
1	Desks gine jate hain (1, 2, 3); baqi naape jate hain.
2	Sirf NAND (ya sirf NOR) se har gate/circuit ban sakta hai.
3	XOR = 1 jab inputs alag: $$A\bar{B} + \bar{A}B$$.
4	Network layer (3): IP addressing + routing.
5	Agile = iterative sprints (2–4 weeks).
6	Beech ka element tabhi half batata hai jab list sorted ho.
7	Nested comparisons: $$\frac{n(n-1)}{2}$$ → $$O(n^2)$$.
8	input() hamesha string deta; maths ke liye int().

Q	Kyun yahi jawab? (MCQ wazahat)
9	// = floor division: 17 = 5×3 + 2 → 3 (int).
10	Chalta hai magar jawab ghalat = logical error.
11	Primary key: unique + not null.
12	Actuator = action (valve kholna); sensor = collect.
13	Wikipedia = khulasa/reference → tertiary.
14	Numbers (naape/gine) = quantitative.
15	Formatted printed output = Report.

Mark the Mock — Section C Worked Solutions

Har long answer ka model · marks ka hisaab

★ **Must lock:** Missed Goldens

C1 — K-map $\sum m(1,3,5,7)$ [10] ★ Golden

A \ BC	00	01	11	10
0	0	1	1	0
1	0	1	1	0

F = C

1s: m1 (001), m3 (011), m5 (101), m7 (111) — sab mein **C = 1**. Ek group of 4 (columns 01 + 11, dono rows). A aur B badalte → gir gaye.

F = C · Logic diagram: input C seedha output Y tak (ek wire / buffer) — koi gate nahi.

Marks: map + Gray headings 3 · sahih 1s 2 · group 2 · expression 2 · diagram 1

C2 — Bubble Sort [5, 1, 4, 2] + Binary Search [10] ★ Golden

Pass	Comparisons	List after
1	5>1 swap · 5>4 swap · 5>2 swap	1, 4, 2, **5**
2	1>4 no · 4>2 swap	1, 2, **4, 5**
3	1>2 no — koi swap nahi	**1, 2, 4, 5** ✔

Step	low	high	mid	data[mid]	Action
1	0	3	1	2	2 < 4 → low = 2
2	2	3	2	4	**Found at index 2**

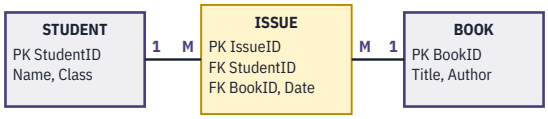
$$mid = \lfloor (0+3)/2 \rfloor = 1$$

, phir

$$\lfloor (2+3)/2 \rfloor = 2$$

.

C4 — Library ER + schema + query [10] ★ Golden



Student(StudentID, Name, Class) · Book(BookID, Title, Author) · Issue(IssueID, StudentID FK, BookID FK, Date)

```
SELECT Book.Title
FROM Issue INNER JOIN Book
  ON Issue.BookID = Book.BookID
WHERE Issue.StudentID = 101;
```

C3 — Average & grade program [10] ★ Golden

```
n = int(input("How many subjects? "))
total = 0
for i in range(n):
    total += float(input("Marks: "))
avg = total / n
print("Average:", avg)
if avg >= 80:
    print("A")
elif avg >= 70:
    print("B")
elif avg >= 60:
    print("C")
else:
    print("D")
```

Sample run: n = 4 → 80, 70, 90, 60 → total 300 →

$$\frac{300}{4} = 75.0$$

→ 75 ≥ 70 → **B**.

Marks: input/loop 3 · average 2 · if-elif ladder sahih order 3 · sample run 2

C5 — Sindh farm: IoT + Analytics + AI; impacts [10]

★ Golden

Soil sensor (IoT) → Cloud sheet of moisture (Analytics) → Model: "too dry" (AI) → Pump ON (actuator)

Education +	Education –
LMS, online lectures — apni speed se parhai	Screen addiction, aankh/kamar dard
Remote classes — door ke students bhi	Shehar/dehaat device & internet gap

Marks: chain + har role 5 · 2 positive 2.5 · 2 negative 2.5

Section C marking guide

- Definition/sahih terms ke baghair → max 6/10.
- Diagram/table bina label → −2.
- Python mein indentation ghalat → −1 to −3.
- Sirf 2 long sawal kiye? Teesre ke 0 — time plan dekho.

C score: ____ / 30

Aam ghaltiyan — Section C (in se marks katte hain)

Q	Ghalti	Sahih tareeqa
C1	Columns 00, 01, 10, 11 likhna	Gray code: 00, 01, **11**, 10
C2	Sirf final list likhna; binary search unsorted list pe	Har pass dikhao; pehle sort, phir low/mid/high table
C3	`input()` ko float/int mein convert na karna; ladder ulta (≥60 pehle)	`float(input())`; sab se sakht condition (≥80) pehle
C4	FK mark na karna; M:N seedha jorna	Issue junction table, dono FK, cardinality 1–M likho
C5	Sirf faide likhna	Chain diagram + 2 positive + 2 negative, Pakistani misaal

Day 29

Score, Repair List & Golden Checklist

90–150 min: repair list bharo · 150–180 min: ek missed Golden yaad se

★ **Must lock:** Missed Goldens

Score tracker

Section	Max	Mine	%
A — MCQ	15		
B — Short	30		
C — Long	30		
Total	**75**		

$$\text{Percentage} = \frac{\text{marks}}{75} \times 100 \qquad \text{e.g. } \frac{54}{75} \times 100 = 72\%$$

Score	Matlab	Kal (Day 30)
≥ 80% (60+)	Tayyar ✔	Sirf repair list, jaldi so jao
60–79% (45–59)	Theek, kuch kamzor	Repair list poori, har item 2 dafa
< 60% (< 45)	Serious repair	Sirf top 6 Goldens — naya kuch nahi

Chapter-wise kamzori

Ch	A ghalat	B kam	C kam	Priority
Ch 1				
Ch 2				
Ch 3				
Ch 4				
Ch 5				
Ch 6				

Repair list ✍

Har Golden jis pe **0 ya aadhe** marks aaye. Kal **sirf yahi** — poori kitaab nahi.

#	Golden topic missed	Main yaad se kya likhunga (def / table / diagram)	Done
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐
9			☐
10			☐

150–180 min: in mein se **ek** Golden abhi yaad se dobara likho.

Golden checklist — mock mein jo miss hua us pe ✔ (45 Goldens)

Chapter	Goldens (miss hua to ✔)	Missed
Ch 1	☐ Analog & Digital · ☐ Logic Gates · ☐ Logic Diagrams · ☐ K-Maps · ☐ SDLC · ☐ Waterfall · ☐ Agile · ☐ OSI 7 layers	__ / 8
Ch 2	☐ Algorithm · ☐ Algo vs Pseudocode · ☐ Decomposition · ☐ Bubble Sort · ☐ Selection Sort · ☐ Binary Search	__ / 6
Ch 3	☐ Python basics · ☐ Variables · ☐ Data Types · ☐ Arithmetic · ☐ Relational/Logical · ☐ Bitwise · ☐ Control Structures · ☐ if / if-else · ☐ if-elif-else · ☐ for & range · ☐ while/break/continue · ☐ Nested loops	__ / 12
Ch 4	☐ Keys & Integrity · ☐ ER Model · ☐ Referential Integrity · ☐ Access Queries	__ / 4
Ch 5	☐ AI · ☐ IoT & Components · ☐ Information Sources · ☐ Assistive Tech	__ / 4
Ch 6	☐ Types of Data · ☐ Collection Strategies · ☐ Interviews & Surveys · ☐ Prototype/Obs/Sim · ☐ Primary vs Secondary · ☐ Inquiry case study · ☐ Steps 1–3 · ☐ Steps 4–6 · ☐ Steps 7–9 · ☐ Steps 10–12 · ☐ Final Artefact	__ / 11

Day **30**

Repair Day — Phir Ruk Jao

Sirf Day 29 ki repair list · 6 pm stop · raat 10 baje neend

★ **Must lock:** Missed Goldens only

Aaj ka plan (3 ghante · 6 baje band)

Waqt	Kaam
0–10′	Day 29 ki repair list kholo — sirf woh items
10–100′	Har missed Golden pe **3-step redo** (neeche)
100–115′	Break — paani, thora chalo; phone nahi
115–170′	Doosra round: **kitaab band**, har item ounchi awaaz mein + ek table/diagram
170–180′	Bag pack — bas! **6 pm stop**

⚠ **Dhoka:** Naya chapter nahi · naya non-Golden card nahi · aakhri raat "sab kuch dobara" nahi.

3-step redo (har missed Golden)

1. Definition bolo (word-for-word) → 2. Table / diagram yaad se banao → 3. Ek 5-mark answer likho

#	Golden	Def	Table/Diag	5-mark
1		☐	☐	☐
2		☐	☐	☐
3		☐	☐	☐
4		☐	☐	☐
5		☐	☐	☐
6		☐	☐	☐
7		☐	☐	☐
8		☐	☐	☐

Raat ki checklist 🎒

☐ Admit card / roll no. slip · ☐ 2 blue/black pens + spare · ☐ Pencil, sharpener, eraser · ☐ Ruler (diagrams, tables) · ☐ Calculator (agar allowed) · ☐ Paani ki bottle · ☐ Ghari (hall clock pe bharosa nahi) · ☐ Kapre tayyar · ☐ Alarm set · ☐ Raat 10 baje so jao 😴

Goldens **dimagh mein** le kar jao — booklet hall mein nahi.

Hall mein — 6 sunehri usool

1. Pehle 10′ poora paper parho; Section C ke apne **3 strongest** sawal chuno.
2. MCQ pehle — jaldi, magar har option parh ke.
3. Har long answer: **Define · Explain · Example · Diagram · Working**.
4. Diagrams pencil + ruler se, **labels aur title** ke saath.
5. Python: indentation, colons, quotes check.
6. Aakhri 10′ — roll no., question numbers, khali chhore sawal check.

5-mark frame — ek nazar 🧠

Step	Kya likho	Misaal (Bubble Sort)
Define	Kitaab wali exact English definition	Bubble Sort repeatedly compares adjacent elements and swaps them if they are in the wrong order.
Explain	2–3 lines Urdu soch, English likhai	Har pass mein sab se bari value aakhir tak; N−1 passes.
Example	Pakistani / rozmarra misaal	Merit list ke marks tarteeb mein.
Diagram	Labelled diagram ya table	Pass-by-pass table (Pair · Compare · Action · List after).
Working	Steps / trace / formula	[8,4,1,9,3] → 4,1,8,3,9 → … → 1,3,4,8,9; comparisons $\frac{N(N-1)}{2}$.

Aakhri nazar — 6 pm se pehle ek dafa 👀

Topic	Aakhri nazar (mnemonic / formula)
OSI 7	Please Do Not Throw Sausage Pizza Away (1→7)
SDLC	R-D-I-T-D-M
Truth table	rows $= 2^n$; K-map Gray 00, 01, 11, 10
CT pillars	Don't Pick Any Apple
Algorithm	S-O-R-A-B
Binary search	$MID = \lfloor \frac{BEG+END}{2} \rfloor$, sorted zaroori

Topic	Aakhri nazar (mnemonic / formula)
Python ops	/ float · // floor · % baqi · $\sim x = -(x+1)$
Loops	I-T-U; range ek pehle ruke
Keys	PK unique + not null; FK → doosri PK
IoT	Sensor senses, actuator acts — 6 components
Assistive	My Small Cat Cries · AII
Ch 6	I Saw People On Screens · C-NAPE · Shows/However/Therefore

🧠 **Yaad rakho:** 30 din mukammal. Tum ne har Golden kam az kam 3 dafa dekha hai. Ab **aaram** — kal behtareen do. 🌟 Best of luck!